

eCOA User Requirements Specification (URS)

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1. INTRODUCTION AND PURPOSE

CureHHT eCOA solution is composed of a sponsor portal and a mobile application with available web-based interface, which are integrated with each other and serve as the primary user interfaces for recording of nosebleed events and completion of participant questionnaires for the TER-4480-C01 study.

The Sponsor Portal is the web-based interface used by clinical staff — Study Coordinators, Clinical Research Associates, and Administrators — to manage participant linking, monitor trial progress, distribute questionnaires, and review submitted data. The Mobile Application is the participant-facing interface used to record daily nosebleed events and complete questionnaires sent from the Sponsor Portal. The two components are integrated in such that data captured in the Mobile Application is synchronized to the Sponsor Portal and onward to Rave EDC, and questionnaires initiated in the Sponsor Portal are delivered to the participant's device.

2. DOCUMENT REVISION HISTORY

Version	Date	Author	Description of Changes
1.0	25-Mar-2026	Elvira Koliadina	Initial draft

3. SIGNATURES

Trial Sponsor Agreement

By my signature below I confirm that the settings have been reviewed and approved within the Trial Sponsor organization.

Approver	Signature
First Name Last Name Study Manager	
First Name Last Name Quality Assurance	
First Name Last Name Data Management	
Print Name & Title	Signature / Date

CureHHT Personnel Agreement

By my signature below I confirm the information contained herein will be implemented as appropriate by CureHHT on behalf of the Trial Sponsor.

Approver	Signature
First Name Last Name Lead Developer	
Elvira Koliadina Project Manager	
Print Name & Title	Signature / Date

4. SYSTEM-WIDE FUNCTIONAL STANDARDS AND CONTROLS

This section defines system-wide functional standards, controls, and behaviors that apply across the entire solution, including the sponsor portal and mobile application, where applicable.

The requirements outlined in this section are global in nature and apply to all system features unless explicitly stated otherwise. The intent of this section is to ensure consistent interpretation, implementation, and use of common system functionality, and to avoid unnecessary repetition within individual, feature-specific requirements.

All functional requirements described in subsequent sections shall adhere to the standards and controls defined herein.

4.1 Common User Interface Elements and Actions

The system shall use a consistent set of commonly displayed elements, buttons, and interaction patterns across the Sponsor Portal and Mobile Application. Where these terms are used in subsequent requirements, their behavior shall be interpreted according to the definitions provided in this section, unless specified otherwise.

Buttons

Element	Behavior
Confirm	Confirms a pending action displayed in a dialog. Proceeds with the action upon click.
Cancel	Discards any unsaved changes and returns the user to the previous state. No data is modified.
OK	Acknowledges a system message or notification and dismisses it. No action is confirmed or cancelled.
Submit	Finalizes and sends the current form or questionnaire data. Once submitted, the action cannot be undone unless explicitly permitted by the relevant requirement.
Next	Advances to the next step in a sequence. Data entered in the current step is preserved.
Back	Returns to the previous step in a sequence. Previously entered data is preserved.
Continue	Resumes a flow that was interrupted or paused. Returns the user to the point where they left off.

More details	Opens a detailed view displaying the full record associated with the current row. Represented as a button or link affordance on the row.
Close	Dismisses a modal or dialog without performing any action. No data is modified.

UI Elements

Element	Behavior
Tab	A navigation element that filters a list by a category.
Search Input	A single text input that filters a list in real time across specified fields.
Status Badge	An inline label displaying the current status of a record.

Actions

Element	Behavior
Edit	Opens an existing record for modification. Changes are not applied until confirmed or submitted. Represented by a pencil icon.

Interaction Patterns

Element	Behavior
Real-time Filtering	A list updates as the user types without requiring explicit submission.
Preserved Search State	The search input is retained when the user navigates between tabs.

Dialog Patterns

Element	Behavior
Acknowledgement Dialog	Displays a message informing the user of a system event or outcome. Requires the user to click OK to dismiss. No action is cancelled or confirmed.
Confirmation Dialog	Displays the consequence of a pending action. Requires the user to either confirm or cancel before the action is applied.

Reason Dialog — Free Text	Displayed before a high-impact action. Requires the user to enter a free text reason of up to 100 characters before proceeding. The action SHALL NOT proceed until a reason is provided.
Reason Dialog — Predefined List	Displayed before a high-impact action. Requires the user to select a reason from a sponsor-defined list before proceeding. The action SHALL NOT proceed until a reason is selected.

4.2 Status Registry

Sponsor Portal User Status

The system maintains a defined set of status values for Sponsor Portal user accounts.

Status	Description
Pending Activation	The account has been created but the user has not yet accepted the invitation or set up their credentials.
Active	The account has been created and the user can log in and perform actions per their role and permissions.
Inactive	The account has been deactivated and the user can no longer log in.

Participant Status

The system maintains a defined set of status values representing the participant's linking, trial participation, device, and data synchronization state.

Status	Description
Not Connected	The participant record has been synced from Rave EDC to the Sponsor Portal but the mobile application has not been linked to the Sponsor Portal.
Pending	An invitation has been sent to the participant but the participant has not yet linked their device.
Expired	An invitation has been sent to the participant but expired before the participant linked their device.
Linked - Awaiting Start	The participant's device is linked to the Sponsor Portal but Diary Data Synchronization has not yet been activated.
Trial Active	Diary Data Synchronization has been activated and diary entries are being transmitted to the Sponsor Portal and Rave EDC.
Disconnected	A Study Coordinator has manually disconnected the participant in the Sponsor Portal. Sponsor-specific rules remain applied to the mobile application but data is not syncing. Data will resume syncing when the participant is reconnected.
Not Participating	The participant has been marked as not participating following disconnection. Sponsor-specific rules are no longer applied to the mobile application.

Participant Status Transitions

The following transitions are valid. All other transitions are prohibited.

From	To	Trigger
Not Connected	Pending	Study Coordinator generates linking code
Pending	Linked - Awaiting Start	Participant successfully enters linking code for the first time
Pending	Expired	The linking code expires as the participant did not use it
Expired	Pending	Study Coordinator regenerates linking code
Linked - Awaiting Start	Trial Active	Study Coordinator starts trial
Linked - Awaiting Start	Disconnected	Study Coordinator disconnects participant
Trial Active	Disconnected	Study Coordinator disconnects participant
Disconnected	Pending	Study Coordinator initiates reconnection
Disconnected	Not Participating	Study Coordinator marks participant as not participating
Not Participating	Disconnected	Study Coordinator reactivates participant

Questionnaire Status

The system maintains a defined set of status values for Questionnaire instances.

Status	Description
Not Sent	No active Questionnaire exists for the current Cycle and a new one may be sent.
Sent	The Questionnaire has been transmitted to the participant's mobile application and is awaiting participant completion.
Delivery Failed	Transmission of the Questionnaire to the participant's mobile application was unsuccessful and the participant has not received it.
Ready to Review	The participant has submitted answers and the Study Coordinator may finalize.
Closed	The Questionnaire for that cycle has been completed by the participant and is finalized by the Study Coordinator. Subsequent Questionnaires for that cycle cannot be generated.

4.3 User Roles and Permissions

This section defines the role model used across the solution: the generic role templates provided by the platform, the inventory of actions subject to access control, and the sponsor portal deployment's role assignments and permissions. All functional requirements that reference roles or permissions depend on the definitions established here.

4.3.1 REQ-p70005: Customizable Role-Based Access Control

The system supports multi-role assignment and seamless role switching without disrupting active sessions. Self-privilege-escalation is prevented structurally by disallowing users to modify role assignments of their own account.

Assertions

- A. The System SHALL allow a **User Account** to be assigned one or more roles.
- B. The System SHALL allow an active role to be switched without terminating the user's session.
- C. The System SHALL NOT permit a user to modify the role assignments of their own **User Account**.

4.3.2 REQ-p07037: Action Inventory

The Action Inventory enumerates every operation subject to role-based access control. It is the authoritative source for the sponsor-level permissions table — every permission granted or denied in a sponsor deployment must reference an action from this inventory. Actions not listed here cannot be granted or denied. Sponsor deployments may extend the inventory with deployment-specific actions.

Definitions

Action: A discrete operation that a user can perform within the System, subject to role-based access control. An Action either changes system state or retrieves protected data.

Action Inventory: The complete enumerated set of Actions the System recognizes, comprising the platform-level inventory in this requirement and any sponsor-level extensions.

Sponsor-Level Action Extension: An additional Action defined by a specific sponsor deployment beyond the platform Action Inventory.

Assertions

- A. The System SHALL enforce role-based access control for every **Action** in the **Action Inventory**.
- B. The System SHALL reject any attempt to perform an **Action** by a user whose active role does not have permission for that **Action**.
- C. The System SHALL permit an **Action** to proceed when the user's active role has permission for that **Action** and the user is within the permitted scope.
- D. The System SHALL support **Sponsor-Level Action Extensions** to the **Action Inventory**.

Action Inventory Table

Category	Action ID	Action
Participant Management	ACT-PAT-001	Link participant
	ACT-PAT-002	Start Trial
	ACT-PAT-003	Disconnect participant
	ACT-PAT-004	Reconnect participant
	ACT-PAT-005	Mark participant as not participating
	ACT-PAT-006	Reactivate participant
	ACT-PAT-007	View participant info
Questionnaire	ACT-QST-001	Send questionnaire
	ACT-QST-002	Call back questionnaire
	ACT-QST-003	Finalize questionnaire

Category	Action ID	Action
User Account	ACT-USR-001	Create User Account
	ACT-USR-002	Edit User Account
	ACT-USR-003	Deactivate User Account
	ACT-USR-004	Reactivate User Account
	ACT-USR-005	Unlock User Account
	ACT-USR-006	Resend activation email
	ACT-USR-007	Assign role to User Account
	ACT-USR-008	Assign Site to User Account
Site	ACT-SIT-001	View Sites
Audit Log	ACT-AUD-001	View audit log
Administrator Settings	ACT-ADM-001	View Administrator Settings

4.3.3 REQ-p03013: Role Definitions

Definitions

Study Coordinator: A portal user responsible for managing day-to-day participant interactions at one or more assigned Sites.

Clinical Research Associate (CRA): A portal user responsible for monitoring Study Coordinator activities and site compliance at one or more assigned Sites.

Administrator: A portal user responsible for managing User Accounts, role assignments, and Site assignments across the study.

Assertions

A. The System SHALL recognize the following roles in the sponsor portal: **Study Coordinator**, **Clinical Research Associate**, and **Administrator**.

B. The System SHALL map the **Study Coordinator** role to the **Investigator** generic role template.

C. The System SHALL map the **Clinical Research Associate** role to the **Auditor** generic role template.

D. The System SHALL map the **Administrator** role to the **Administrator** generic role template.

4.3.4 REQ-CAL-p00016: Permissions Table

The permissions table defines which roles are granted each Action from the Action Inventory (REQ-p07037).

Assertions

A. The System SHALL enforce the following permission rules in the sponsor portal:

User Account Management

Action ID	Action	Study Coordinator	CRA	Administrator
ACT-USR-001	Create User Account	X	X	✓
ACT-USR-002	Edit User Account	X	X	✓
ACT-USR-003	Deactivate User Account	X	X	✓
ACT-USR-004	Reactivate User Account	X	X	✓
ACT-USR-005	Unlock User Account	X	X	✓
ACT-USR-006	Resend activation email	X	X	✓
ACT-USR-007	Assign role to User Account	X	X	✓
ACT-USR-008	Assign Site to User Account	X	X	✓

Participant Management

Action ID	Action	Study Coordinator	CRA	Administrator
ACT-PAT-001	Link participant	✓	X	X
ACT-PAT-002	Disconnect participant	✓	X	X
ACT-PAT-003	Reconnect participant	✓	X	X
ACT-PAT-004	Mark participant as not participating	✓	X	X
ACT-PAT-005	Reactivate participant	✓	X	X
ACT-PAT-006	View participant	✓	✓	X

Questionnaire Management

Action ID	Action	Study Coordinator	CRA	Administrator
ACT-QST-001	Send questionnaire	✓	✗	✗
ACT-QST-002	Call back questionnaire	✓	✗	✗
ACT-QST-003	Finalize questionnaire	✓	✗	✗

Sites and Audit Log

Action ID	Action	Study Coordinator	CRA	Administrator
ACT-SIT-001	View Sites	✓	✓	✓
ACT-AUD-001	View audit log	✓	✓	✓

4.3.5 REQ-CAL-p00087: Site View Scope

Assertions

- A. A **Study Coordinator** SHALL only view **Sites** to which they are assigned.
- B. A **Clinical Research Associate** SHALL only view **Sites** to which they are assigned.
- C. An **Administrator** SHALL view all **Sites** within the sponsor portal.

4.3.6 GUI-p03045: Role Switching — Interface Behavior

Definitions

Role Selector: UI element displayed in the header that shows the user's currently active role and allows users with multiple assigned roles to switch their active role.

Assertions

Display

- A. The interface SHALL display the **Role Selector** in the header for any user assigned two or more roles.
- B. The interface SHALL NOT display the **Role Selector** for a user assigned exactly one role.
- C. The **Role Selector** SHALL display the user's currently active role.
- D. When opened, the **Role Selector** SHALL display the complete list of the user's assigned roles and SHALL visually indicate which role is currently active.

Interaction

- E. When the user selects a role from the **Role Selector**, the interface SHALL set that role as the active role.
- F. When the active role changes, the interface SHALL load the default landing view for the selected role.
- G. The **Role Selector** SHALL NOT present a confirmation step before switching roles.

4.4. Questionnaire Score Display

4.4.1 REQ-CAL-p00007: Questionnaire Score Display Prohibition

The platform does not permit calculated questionnaire scores to be displayed to any user.

Assertions:

A. The System SHALL NOT display calculated Questionnaire Score to any user.

4.5 Audit Log Visibility

This section defines per-role visibility scope of audit log records and the user interface presentation of the **Audit Log View**. The content captured in each **Audit Log Entry**.

4.5.1 REQ-CAL-p00088: Audit Log View Scope

The System maintains an audit trail of all Actions performed by users. Each role requires a different scope of visibility into that audit trail to support its responsibilities while preserving separation of duties. **Study Coordinators** need visibility into their own Actions for self-verification and to support documentation requests during monitoring. CRAs need visibility into **Study Coordinator** Actions at their assigned Sites to perform monitoring and compliance verification. **Administrators** need visibility into **User Account** management Actions to support account lifecycle verification and troubleshooting.

Definitions

Audit Log Entry: A single immutable record capturing one **Action** performed in the System, including the **Action** performed, the user who performed it, the date and time, and any associated context.

Audit Log View: The interface presented to a user that displays a scoped set of Audit Log Entries the user is authorized to see.

Assertions

General

- A. The System SHALL create an **Audit Log Entry** for every **Action** in the **Action Inventory** (REQ-p07037 for reference) that is performed in the System.
- B. The Actions visible in any **Audit Log View** are drawn from the **Action Inventory**.

Study Coordinator Scope

- C. The System SHALL provide each **Study Coordinator** with read-only access to the **Audit Log View**.
- D. The System SHALL include in a **Study Coordinator's Audit Log View** only Audit Log Entries for Actions performed by that **Study Coordinator**.

CRA Scope

- E. The System SHALL provide each **CRA** with an **Audit Log View** scoped to a single **Site** selected by the **CRA** from their assigned Sites.
- F. The System SHALL include in a **CRA's Audit Log View** only Audit Log Entries for Actions performed by Study Coordinators at the selected **Site**.
- G. The System SHALL exclude from a **CRA's Audit Log View** all Audit Log Entries for Sites not assigned to the **CRA**.
- H. The System SHALL provide CRAs with read-only access to the **Audit Log View**.

Administrator Scope

- I. The System SHALL provide each **Administrator** with read-only access to the **Audit Log View**.
- J. The System SHALL include in an **Administrator's Audit Log View** all Audit Log Entries for **User Account** management Actions performed by any **Administrator**.

4.5.2 GUI-p20077: Audit Log View — Common Presentation

The **Audit Log View** uses a consistent presentation across all roles. This requirement defines the shared display, ordering, content, and empty state behavior. Role-specific GUI requirements refine this base by adding role-specific columns, controls, and entry points.

Assertions

Display

- A. The interface SHALL display the **Audit Log View** as a table containing at minimum the following columns: Timestamp, Action, User, Details.
- B. The interface SHALL display Audit Log Entries in reverse chronological order, with the most recent entry first.

Content

- C. When an **Audit Log Entry** corresponds to an **Action** that included a free text input from the user (such as a reason, justification, or note), the interface SHALL include that text in the Details column.
- D. The interface SHALL display in the Details column a human-readable summary of the **Action** that includes, where applicable, the previous state of the affected record and the parameter or option selected as part of the **Action**.

Empty State

- E. When the **Audit Log View** contains no Audit Log Entries within the active scope or filter selection, the interface SHALL display a message indicating no results were found.

Action Display

- F. The interface SHALL display the **Action** name in the Action column.
- G. When an **Action** has parameters that distinguish meaningful variants (such as **Questionnaire Type**), the interface SHALL append the parameter to the **Action** name in the Action column for human readability.

Detail View

- H. The interface SHALL provide a **More details** button on each **Audit Log Entry** that, when selected, displays the full **Audit Log Entry** record as raw text.

4.5.3 GUI-p20074: Study Coordinator Audit Log View

The **Study Coordinator Audit Log View** presents the **Study Coordinator's** own Actions. **Participant**-related Actions are displayed alongside a **Participant ID** column.

Assertions

Display

A. The interface SHALL display the **Participant ID** column in addition to the columns defined in GUI-p20077.

Controls

B. The interface SHALL provide a **Participant ID** search input that filters the displayed Audit Log Entries by **Participant ID** in real time.

4.5.4 GUI-p20075: CRA Audit Log View

The **CRA Audit Log View** is accessed by selecting a **Site** from the **CRA**'s list of assigned Sites and presents **Study Coordinator** Actions for the selected **Site**. A **Study Coordinator** filter and a **Participant ID** search input enable focused review.

Assertions

Display

A. The interface SHALL display the **Participant ID** and **Site** columns in addition to the columns defined in GUI-p20077.

B. The interface SHALL display the name of the selected **Site** in the **Audit Log View** header.

Controls

C. The interface SHALL provide a **Study Coordinator** selector that filters the displayed Audit Log Entries by the selected **Study Coordinator**.

D. The interface SHALL provide a default selection in the **Study Coordinator** selector that displays Audit Log Entries for all Study Coordinators at the selected **Site**.

E. The interface SHALL provide a **Participant ID** search input that filters the displayed Audit Log Entries by **Participant ID** in real time.

4.5.5 GUI-p20076: Administrator Audit Log View

Assertions

A. The interface SHALL present the **Administrator Audit Log View** on a dedicated tab within the **Administrator** dashboard, alongside the **User Account** management tab.

4.6 Notification Behavior

This section defines system-wide standards for **Push Notifications** delivered by the **Mobile Application** to the **Participant's** device. The standards defined herein apply to all notifications generated by the **Mobile Application** and shall not be restated within individual notification requirements.

4.6.1 REQ-p20078: Notification Behavior

Push Notifications delivered by the **Mobile Application** serve as nudges that prompt the **Participant** to open the application.

Definitions

Push Notification: A notification delivered by the **Mobile Application** to the **Participant's** device notification center via the operating system's notification service.

Assertions

Tap Behavior

- A. When the **Participant** taps a **Push Notification** delivered by the **Mobile Application**, the System SHALL open the **Mobile Application** to the **Main Screen**.
- B. The System SHALL NOT include response actions within the body of a **Push Notification**.

Offline Delivery

- C. When the **Mobile Application** is offline at the time a **Push Notification** is generated, the System SHALL deliver the **Push Notification** when the **Mobile Application** next establishes connectivity.

4.7 Help and Resources

4.7.1 REQ-p20081: Help and Resources

The **Sponsor Portal** provides users with access to in-portal help materials including video tutorials, frequently asked questions, an interactive guided tour, and a downloadable user guide. The materials surfaced are role-aware so that each user is presented with content relevant to their responsibilities. The detailed inventory of materials, tour content, and FAQ entries is maintained in internal documentation; this requirement defines only the platform-level expectation that the help affordance exists and is available to all **Sponsor Portal** roles.

Definitions

Help and Resources: The collection of in-portal help materials available to **Sponsor Portal** users, comprising video tutorials, frequently asked questions, an Interactive Tour, and a downloadable user guide.

Interactive Tour: A guided walkthrough of the **Sponsor Portal** interface presented to a user on first login and replayable on demand from **Help and Resources**.

Assertions

Availability

A. The **System** SHALL make **Help and Resources** available to every **Administrator**, **Clinical Research Associate**, and **Study Coordinator** from any screen of the **Sponsor Portal**.

Content

B. The **System** SHALL present **Help and Resources** content scoped to the user's active **Role**.

C. **Help and Resources** SHALL include video tutorials, frequently asked questions, the **Interactive Tour**, and a downloadable user guide.

Interactive Tour

D. The **System** SHALL present the **Interactive Tour** to a user on their first successful login to the **Sponsor Portal**.

E. The **System** SHALL allow a user to replay the **Interactive Tour** on demand from **Help and Resources**.

4.8 Configuration

4.8.1 REQ-p70020: Configuration Precedence

The platform defines a baseline set of behaviors and default values that apply to every deployment. Each sponsor deployment may, where the platform requirement explicitly permits configuration, override those defaults with values appropriate to that study's protocol. This requirement establishes the precedence rule that resolves any apparent conflict between a platform requirement and a sponsor configuration: when both apply to the same behavior, the sponsor configuration governs. The rule is structural — it ensures that sponsor-specific values such as thresholds, reason lists, reminder times, and enabled features take effect deterministically without requiring each individual platform requirement to restate the precedence relationship. The rule applies only within the bounds set by the platform requirement itself; sponsor configuration cannot grant capabilities the platform does not expose, and cannot override platform behavior that is not declared as configurable.

Definitions

Platform Requirement: A requirement that defines behavior, structure, or default values applicable to every deployment of the **System**. Identified by a requirement ID of the form REQ-pXXXXX or GUI-pXXXXX.

Sponsor Configuration Requirement: A requirement that specifies the values applied to a configurable parameter exposed by a **Platform Requirement** for a specific sponsor deployment. Identified by a requirement ID of the form REQ-CAL-pXXXXX for the sponsor-specific deployment, with analogous prefixes for other deployments.

Configurable Parameter: A value, threshold, list, label, enablement flag, or option that a **Platform Requirement** explicitly declares as configurable per deployment.

Configuration Conflict: A state in which a **Sponsor Configuration Requirement** specifies a value for a **Configurable Parameter** that differs from the default value or behavior described in the corresponding **Platform Requirement**.

Assertions

A. The **System** SHALL apply the value specified in a **Sponsor Configuration Requirement** in place of the default value or behavior described in the corresponding **Platform Requirement** when a **Configuration Conflict** exists.

B. The **System** SHALL apply a **Sponsor Configuration Requirement** only to a **Configurable Parameter** that is explicitly declared as configurable by a **Platform Requirement**.

- C. The **System** SHALL NOT permit a **Sponsor Configuration Requirement** to override behavior of a **Platform Requirement** that is not declared as configurable.
- D. The **System** SHALL NOT permit a **Sponsor Configuration Requirement** to grant a capability that is not defined by a **Platform Requirement**.
- E. When no **Sponsor Configuration Requirement** exists for a given **Configurable Parameter**, the **System** SHALL apply the default value or behavior described in the corresponding **Platform Requirement**.
- F. When a **Platform Requirement** specifies that a behavior takes effect only if a corresponding **Configurable Parameter** is configured, and no **Sponsor Configuration Requirement** configures it, the **System** SHALL NOT apply that behavior.

5. SPONSOR PORTAL REQUIREMENTS

Sponsor Portal: The web-based interface provided by the System for clinical staff and Sponsor administrators to manage User Accounts, Participants, Questionnaires, and audit logs. The **Sponsor Portal** is one of the two primary system surfaces; the other is the **Mobile Application**.

5.1 Administration

5.1.1 REQ-p20029: Create User Account

Definitions:

User Account: The complete set of identity, credential, and authorization data that the **Sponsor Portal** maintains for a single human user.

Full Name: A common name, which is **PII**, of the person associated with a **User Account** that identifies a specific individual in the context of the **Sponsor Portal**.

Email Address: A unique technical identifier used as a destination for system notifications and as a username for authentication to access the system.

Assertions:

- A. The **System** SHALL require a **Full Name**, **Email Address**, at least one **Role**, and — for non-**Administrator** accounts — at least one **Site** to be selected before a **User Account** is created.
- B. The **System** SHALL validate the format of the email address provided (e.g., presence of an “@” symbol).
- C. The **System** SHALL allow assignment of the **Administrator**, **CRA** and **Study Coordinator** role during User Account creation.

5.1.2 REQ-p20035: Account Activation Workflow

The account activation process starts when a user account is created, reactivated, or when an activation email is sent again.

Activation links are time-limited and can only be used once, which helps protect account security. A user cannot access the system until their account has been successfully activated.

Definitions:

Verification Link: A unique, time-limited, single-use secured URL delivered to an email address to confirm that the recipient controls that address.

Activation Webpage: A **Verification Link** associated with a **User Account** that provides a mechanism for the **Account Owner** to activate their account and configure their password and 2FA.

Assertions:

Account States

A. When the status of the **User Account** is **Pending Activation**, the user SHALL NOT be able to access the system.

Activation Webpage Lifecycle

B. When a **User Account** is set to **Pending Activation**, the **System** SHALL create a unique **Activation Webpage** for that **User Account**.

C. When an **Activation Webpage** is created, the **System** SHALL deliver it to the associated **User Account** owner's registered **Email Address**. The username of the user is the **Email Address** that has been entered during user creation.

D. When a **User Account** owner has activated their account and configured a valid password and 2FA, the **System** SHALL set its status to **Active**.

Email Address Verification Lifecycle

E. When an **Administrator** updates the email address of a user, a new **Verification Link** is generated and sent to that updated email address., When the **Verification Link** is used, the **System** SHALL replace the previous **Email Address** with the new **Email Address** as the **User Account's** username.

Verification Link Behavior

F. When a **Verification Link** is used or 14 days have passed from generation, whichever occurs first, the **System** SHALL invalidate the **Verification Link**.

G. When a new **Verification Link** is issued for a **User Account**, the **System** SHALL invalidate any previously issued **Verification Link** for that **User Account**.

Rejection Behavior

H. When a user attempts to access an invalidated **Verification Link**, the **System** SHALL reject the attempt and display the following message: "This link is no longer valid. Please contact your Administrator to request a new activation email."

5.1.3 REQ-p20034: Site Assignment

Sites are managed in **Rave EDC**. The portal displays **sites** for assignment purposes only and does not allow modification to maintain data integrity. Near real-time synchronization ensures that the site list reflects the data within **Rave EDC** at all times.

Definitions

Site: A clinical research location authorized by the Sponsor to enroll **Participants** and conduct trial activities. Each Site is uniquely identified and synchronized from Rave EDC.

Assertions

- A. The system SHALL include only **Sites** that are created in **Rave EDC**.
- B. The system SHALL refresh the list of valid **Sites** each time an **Administrator** logs in.
- C. An **Administrator** SHALL have access to all **Sites**.
- D. The system SHALL allow an **Administrator** to assign any **Site** to a **User Account**.

5.1.4 REQ-p20033: Resend Activation Email

A user may not receive or may lose their activation email before completing setup. Resending generates a new **Activation Webpage** and invalidates the previous one, ensuring only one valid link exists at any time.

Assertions:

- A. While the status of the **User Account** is **Pending Activation**, the system SHALL allow the **Administrator** to resend a new activation email.
- B. The system SHALL NOT impose a limit on the number of resend attempts for a **Pending Activation User Account**.
- C. When an activation email is resent, the system SHALL create a new **Activation Webpage** for that **User Account** and invalidate the previous one.

5.1.5 REQ-p20030: Edit User Account

This functionality allows user access to be updated as roles or responsibilities change. If access is removed or changed, the user is immediately logged out by the system to prevent access to information they should no longer see. When an email address is updated, the new one must be verified by the user again before using his user account.

Assertions:

- A. The **System** SHALL allow an **Administrator** to edit any **User Account** other than their own.
- B. The **System** SHALL require at least one **Role** for **non-Administrator** users.
- C. The **System** SHALL require at least one **Site** for **non-Administrator** users.
- D. The **System** SHALL validate **Email Address** uniqueness across all existing **User Accounts**.
- E. When an **Administrator** changes a **User Account's Role** or **Site**, the **System** SHALL enforce the change immediately.
- F. When an **Administrator** changes a **User Account's Email Address**, the **System** SHALL deliver a **Verification Link** to the new **Email Address**.
- G. When an **Administrator** initiates an **Email Address** change, the **System** SHALL notify the **User Account** owner at the original **Email Address** that a change has been initiated.

5.1.6 REQ-p20031: Deactivate User Account

Deactivation — The action of revoking a **user account's** ability to access the system without permanently removing the account or its associated data. An **inactivated** account retains all historical data and **audit trail** but cannot be used to log in or maintain active sessions.

Assertions:

- A. The System SHALL allow the Administrator to deactivate any **user account** except their own.
- B. The System SHALL terminate all active sessions associated with a **user account** immediately upon **deactivation**.
- C. The System SHALL prevent login to a **deactivated user account**.
- D. The System SHALL preserve all data and **audit trail** associated with a **deactivated user account**.
- E. The System SHALL set the status of a **deactivated user account** to "**Deactivated**".
- G. When an **Administrator** deactivates a **User Account**, the **System** SHALL require a free text reason before applying the change. The reason SHALL be captured in the audit trail.

5.1.7 REQ-p20032: Reactivate User Account

Assertions:

- A. The System SHALL support reactivation of any deactivated **User Account**.
- B. When a **User Account** is reactivated, the system SHALL restore all previously assigned roles and **site** assignments.
- C. Upon reactivation, the status of the **User Account** SHALL be set to **Pending Activation** state.
- D. When an **Administrator** reactivates a **User Account**, the **System** SHALL require a free text reason before applying the change. The reason SHALL be captured in the audit trail.

5.2 User Interface

5.2.1 REQ-p20066: Reason Field Constraints

Where a reason is required before an action proceeds, consistent input constraints ensure reasons are meaningful and auditable. Empty or whitespace-only submissions are rejected to ensure every high-impact action is accompanied by a documented reason for compliance purposes.

Assertions:

- A. Where the **System** requires a reason, the **System** SHALL reject a submission where the reason is empty or contains only whitespace.
- B. Where the **System** requires a reason, the **System** SHALL enforce a maximum input length of 100 characters.

5.2.2 GUI-p00067: User Management Tabs

The user management interface separates active and inactive accounts into distinct tabs to reduce cognitive load and prevent accidental actions on the wrong account. Real-time search and preserved search state allow administrators to work efficiently across tabs without losing context.

Definitions:

User Information Modal : A modal dialog displaying the details and available actions for a selected **User Account**.

Assertions:

Tab Display

- A. The interface SHALL display two tabs: Active Users and Inactive Users.
- B. The Active Users tab SHALL display all User Accounts with a status of Active or Pending Activation.
- C. The Inactive Users tab SHALL display all User Accounts with a status of Deactivated.
- D. The interface SHALL display the Active Users tab by default.
- E. The interface SHALL highlight the currently active tab.
- F. Each tab SHALL display the total count of User Accounts it contains inline with the tab label.
- G. When a User Account status changes, the interface SHALL move the account to the appropriate tab immediately.

Search Behavior

- H. The interface SHALL provide a single search input that filters User Accounts by Full Name and Email Address.
- I. The interface SHALL update search results in real time as the user types.
- J. The interface SHALL preserve the search state when the user switches between tabs.

Empty State

K. When a tab contains no User Accounts matching the current search, the interface SHALL display a message indicating no results were found.

L. The interface SHALL display each **User Account** as a row showing **Full Name**, **Email Address**, **Role(s)**, **Assigned Sites**, and **Status**.

M. Each **User Account** row SHALL be selectable and open the **User Information Modal**.

N. The interface SHALL display a **Create User** action.

5.2.3 GUI-p00031: Deactivate User Account

The deactivation action is available from the Active Users tab only, since deactivated accounts are no longer present there after the action completes.

Assertions:

Availability

A. The interface SHALL make the deactivation action available for every **User Account** displayed in the **Active Users** tab, with the exception of the current user's own account.

Interaction Flow

B. When the deactivation action is initiated, the interface SHALL display the confirmation dialog and require a free text reason before proceeding.

C. The interface SHALL not allow the deactivation action to proceed until a reason has been provided.

D. If the user cancels the confirmation, the **User Account** will remain unchanged.

E. When deactivation is confirmed, the interface SHALL move the **User Account** to the **Inactive Users** tab immediately.

5.2.4 GUI-p00032: Reactivate User Account

Reactivation is initiated from the Inactive Users tab and immediately returns the account to the Active Users tab with Pending Activation status, requiring the user to complete the activation workflow before regaining access.

Assertions:

Availability

A. The interface SHALL make the reactivation action available for every **User Account** displayed in the **Inactive Users** tab.

Interaction Flow

B. When the reactivation action is initiated, the interface SHALL display the confirmation dialog and require a free text reason before proceeding.

C. The interface SHALL NOT allow the reactivation action to proceed until a reason has been provided.

D. If the user cancels the confirmation, the **User Account** will remain unchanged.

E. When reactivation is confirmed, the interface SHALL move the **User Account** to the **Active Users** tab immediately with a status of **Pending Activation**.

5.2.5. GUI-p00033: User Information Modal

The User Information Modal provides a summary view of a User Account and the primary actions available for that account. All account management actions are initiated from this modal.

Assertions

Display

- A. The User Information Modal SHALL display the **Full Name**, **Email Address**, **Status**, assigned **Role(s)**, and **Site** assignments of the selected **User Account**.
- B. The User Information Modal SHALL display every **Role** assigned to the **User Account**.
- C. When the **User Account** is an **Administrator**, the interface SHALL indicate that the **User Account** has access to all **Sites**.
- D. When the **User Account** holds one or more non-**Administrator Roles**, the interface SHALL allow the viewer to select a single **Role** to scope the **Site** list.
- E. When a **Role** is selected, the interface SHALL display only the **Site(s)** assigned to the **User Account** under that **Role**.
- F. The interface SHALL display the total count of **Sites** assigned to the selected **Role**.
- G. The interface SHALL display each **Site** entry using the **Site** identifier and **Site** name.
- H. The interface SHALL indicate which **Role** is currently selected.

Actions

- I. The **User Information Modal** SHALL present an **Edit User** action, a **Deactivate User** action, and a **Close** action for accounts displayed in the **Active Users** tab.
- J. The **User Information Modal** SHALL present a **Reactivate User** action and a **Close** action for accounts displayed in the **Inactive Users** tab.
- K. The **User Information Modal** SHALL NOT present a **Deactivate User** action for the currently authenticated user's own account.
- L. When the user selects **Close**, the interface SHALL dismiss the modal without making any changes.

M. When the user selects **Edit User**, the interface SHALL open the edit workflow for that **User Account**.

5.2.6 GUI-p20080: Administrator Dashboard

The Administrator Dashboard is the primary surface for an Administrator to manage **User Accounts** and review audit log activity. Organising the surface as a container with two top-level tabs separates User Account management from audit log review while keeping both areas reachable from a single entry point. The dashboard scaffolding defined here establishes the header, the top-level tab navigation, and the default landing tab; the contents of each tab are governed by the requirements referenced from those tabs.

Definitions

Administrator Dashboard: The default surface presented to an **Administrator** upon successful authentication to the **Sponsor Portal**, containing top-level navigation to **User Account** management and audit log review.

Assertions

- A. The interface SHALL display a header at the top of the **Administrator Dashboard** containing the **Sponsor Portal** name, the **Full Name** of the authenticated user, the user's active **Role**, a **Settings** action, and a **Logout** action.
- B. The header SHALL remain visible on every screen of the **Administrator Dashboard**.
- C. When the **Administrator** selects **Settings**, the interface SHALL navigate to the **Administrator Settings** surface defined in GUI-p20083.

Top-Level Tabs

- D. The **Administrator Dashboard** SHALL display two top-level tabs: **Users** and **Audit Logs**.
- E. The **Users** tab SHALL display the User Management interface defined in GUI-p00067.
- F. The **Audit Logs** tab SHALL display the Administrator Audit Log View defined in GUI-p20076.
- G. The interface SHALL display the **Users** tab by default upon login.
- H. The interface SHALL display a visual indicator identifying the currently active top-level tab.

Logout

- I. When the **Administrator** selects **Logout**, the **System** SHALL terminate the **Session** and return the **Administrator** to the **Sponsor Portal** login interface.

5.3 Questionnaire Management

5.3.1 REQ-p01065: Clinical Questionnaire System

Clinical trials use structured questionnaires to collect reliable data in a consistent and compliant way. The questionnaires are carefully built to ensure accuracy and a good participant experience. New questionnaires can be added as needed, and the content always comes from approved documents, not edited directly in the app.

Definitions:

Questionnaire: A validated data collection instrument presented to a participant to capture self-reported clinical outcomes at protocol-defined points in a clinical trial.

Questionnaire Type: A named category of Questionnaire. Each Questionnaire Type is implemented as an individual coded component.

Trial Start: The event that formally begins a participant's active trial participation. The trigger and label for Trial Start are sponsor-configurable.

Completion Status: The current state of a Questionnaire instance for a given participant. Valid values are defined per Questionnaire workflow.

Diary Data Synchronization: The continuous transmission of participant diary entries from the participant's mobile application to the Sponsor Portal and Rave EDC.

Questionnaire Display Name: The participant-facing name of a Questionnaire as presented in the mobile Application. Distinct from the instrument name used for internal identification and clinical reference.

Trial: A clinical study conducted under a Sponsor's protocol in which Participants are enrolled to capture self-reported clinical outcomes through the Mobile Application. The Trial begins at Trial Start.

Assertions:

Questionnaire Registry

- A. The System SHALL allow implementation of multiple **Questionnaire Types**, each as an individual coded component.
- B. The System SHALL track **Completion Status** for each **Questionnaire** instance.

Data Synchronization

- C. The System SHALL activate **Diary Data Synchronization** upon **Trial Start**.
- D. The System SHALL deactivate **Diary Data Synchronization** when a participant is disconnected or is marked as Not Participating.

Configuration

E. The System SHALL support sponsor-configurable configuration of the trigger and workflow that activates **Trial Start**.

F. Each Questionnaire SHALL have a Questionnaire Display Name.

5.3.2 REQ-p01053: Sponsor Questionnaire Configuration

A configuration-driven approach allows the sponsor to tailor questionnaire behavior to their study protocol without requiring platform code changes, ensuring that mid-study modifications are managed deliberately and do not unintentionally affect data collection.

Assertions:

- A. The System SHALL support sponsor-specific configuration of which **Questionnaires** are enabled per deployment.
- B. The System SHALL support sponsor-specific configuration of participant reminder behavior for each enabled **Questionnaire**.
- C. The System SHALL support sponsor-specific configuration of the processing steps that occur between participant submission and Rave EDC synchronization for each **Questionnaire**.
- D. The System SHALL support sponsor-specific configuration of whether a participant may edit their answers after submission and under what conditions.
- E. The System SHALL support sponsor-specific configuration of which languages are enabled per **Questionnaire**.

5.3.3 REQ-p01050: Clinical Data Submission Control

Clinical data integrity requires that the system only accepts data conforming to approved, predefined structures. A controlled registry of approved data collection instruments ensures that unrecognized or malformed data is rejected at the point of entry, supporting audit trail integrity and validation requirements under 21 CFR Part 11.

Assertions:

- A. The System SHALL maintain a controlled list of approved **Questionnaire** definitions.
- B. The System SHALL reject any data submission that does not conform to an approved **Questionnaire** definition.

5.3.4 REQ-p02047: Questionnaire Change Control

Questionnaires must stay the same throughout an active trial so the data remain consistent and traceable. Any updates must follow a formal change process to protect the integrity of the system.

Assertions:

- A. The System SHALL use predefined **Questionnaires** for all clinical data collection.
- B. The System SHALL NOT allow **Questionnaires** to be created, modified, or deleted during trial operation. If any new **Questionnaire** is required, a new **Questionnaire Type** will be implemented. Modifications for an existing **Questionnaire** will follow the process outlined in the below two assertions.
- C. Changes to **Questionnaires** SHALL be subject to formal change control procedures before deployment.
- D. The System SHALL maintain backward compatibility to allow viewing of data collected with previous **Questionnaire** versions.

5.3.5 REQ-CAL-p00022: Start Trial Workflow

Before the trial is started, the diary entries remain on the participant's device and are not transmitted to the Sponsor Portal or Rave EDC.

Definitions:

Start Trial: The action available to a Study Coordinator for a participant with Linked - Awaiting Start status that activates Diary Data Synchronization and begins the participant's active trial participation.

Assertions:

- A. When a Study Coordinator confirms **Start Trial** for a participant with **Linked - Awaiting Start** status, the System SHALL change the Participant Status to **Trial Active** and activate **Diary Data Synchronization**.
- B. The System SHALL NOT activate **Diary Data Synchronization** for a participant before **Start Trial** is confirmed.
- C. The System SHALL NOT permit **NOSE HHT** or **HHT-QoL** Questionnaires to be sent to a participant before **Start Trial** is confirmed.

5.3.6 GUI-CAL-p00005: Start Trial Workflow Interface

Assertions:

- A. The interface SHALL present a **Start Trial** button to the Study Coordinator in the Action column for participants with **Linked - Awaiting Start** status.
- B. When the Study Coordinator selects **Start Trial**, the interface SHALL display a Confirmation Dialog.
- C. The Confirmation Dialog SHALL present a **Send EQ** button and a cancel action.
- D. When the Study Coordinator selects **Send EQ**, the interface SHALL update the participant's status to **Trial Active** and replace the **Start Trial** button with a **Manage Questionnaires** button in the Action column.
- E. If the Study Coordinator cancels the Confirmation Dialog, the participant's status SHALL remain unchanged.

5.3.7 REQ-CAL-p00023: NOSE HHT and HHT-QoL Questionnaire Workflow

NOSE HHT and HHT-QoL questionnaires are administered to participants on Day 1 of each treatment cycle. A Study Coordinator initiates each cycle by sending the questionnaire(s) to the participant's mobile application.

Assertions:

Sending

- A. The System SHALL enforce a limit of one active **Questionnaire** per **Cycle** per **Questionnaire Type**.
- B. When a **Questionnaire** fails to reach the participant's mobile application, the System SHALL change its status to **Delivery Failed**.
- C. After the first **Cycle** has been finalized, the System SHALL make the **Start Next Cycle** available in place of **Send Now** action for that **Questionnaire Type**.

Deleting

- D. The System SHALL allow a Study Coordinator to call back a **Questionnaire** with **Sent**, **Delivery Failed** or **Ready to Review** status.
- E. The System SHALL NOT allow a Study Coordinator to call back a **Questionnaire** when it is already completed by the participant and finalized by the Study Coordinator. .

Participant Completion

- F. The System SHALL allow the participant to edit their answers at any time before Finalization (i.e., while the Questionnaire has Sent or Ready to Review status). There is no limit on how many times a participant may edit their answers.
- G. When a participant submits a **Questionnaire**, the System SHALL change its status to **Ready to Review**.

Site Visit Review

- H. The System SHALL allow a Study Coordinator to review submitted **Questionnaire** answers before finalization.

Finalizing

- I. The System SHALL allow a Study Coordinator to finalize a **Questionnaire** with **Ready to Review** status.
- J. Upon finalization, the System SHALL lock the answers, calculate the **Questionnaire** score, and transmit answers to Rave EDC.
- K. When the finalization operation fails, the System SHALL automatically retry the operation in the background. The **Questionnaire** SHALL remain in **Ready to Review** status throughout.

L. While a finalization retry is in progress, the System SHALL display the following message to the Study Coordinator: "Submission is taking longer than expected. The system is retrying. No action is required."

M. Upon successful finalization, the System SHALL change the **Questionnaire** status to **Not Sent**.

5.3.8 GUI-CAL-p00006: Manage Questionnaires Modal

Definitions:

Questionnaire Card: The UI component displayed within the Manage Questionnaires Modal representing a single Questionnaire Type for a participant, showing its current status and available actions.

Select Starting Cycle Dialog: The dialog displayed when a Study Coordinator selects Send Now for the first time for a given Questionnaire Type, allowing selection of the Starting Cycle before sending.

Assertions:

Modal

A. The interface SHALL display the Manage Questionnaires Modal with the participant ID in the modal header.

B. The interface SHALL display one **Questionnaire Card** per **Questionnaire Type**.

C. The interface SHALL present a close action that dismisses the modal without making any changes.

Questionnaire Card — Display and Actions

D. Each **Questionnaire Card** SHALL display the **Questionnaire Type** name and the current **Questionnaire Status**. Date and time information SHALL be paired inline with the **Cycle** it describes (e.g., "Cycle 1 · Apr 24, 2026, 11:34 AM"), not as a standalone field.

E. The interface SHALL present cycle information, **Status Badge** placement, and available actions on each **Questionnaire Card** according to the following table:

Questionnaire Status	Cycle Fields Displayed	Status Badge Placement	Actions Available
Not Sent — never sent	None	Below Questionnaire Type name	Send Now
Not Sent — after finalize or call back	Finalized Cycle (with date), Next Cycle	Inline with Finalized Cycle row	Start Next Cycle
Sent	Current Cycle	Inline with Current Cycle row	Call Back (Trash icon)
Delivery Failed	Current Cycle	Inline with Current Cycle row, with information icon (i)	Finalize, Call Back (Trash icon)

Ready to Review	Current Cycle (with completion date)	Inline with Current Cycle row	Finalize, Call Back (Trash icon)
Closed	Finalized Cycle (with date)	Inline with Finalized Cycle row, as combined badge ("Closed · End of Treatment" or "Closed · End of Study")	None

Delete Interaction

F. When a Study Coordinator selects the call back (Trash icon)action, the interface SHALL display a Reason Dialog — Free Text before proceeding.

G. When the Study Coordinator confirms the call back action, the interface SHALL update the **Questionnaire Card** status to **Not Sent**.

H. When the Study Coordinator cancels the Reason Dialog, the **Questionnaire** SHALL remain unchanged.

Send Now

I. The **Send Now** action is displayed only when the first Questionnaire of each Questionnaire Type will be sent to the participant. When a Study Coordinator selects **Send Now**, the interface SHALL display the **Select Starting Cycle Dialog**.

J. The **Select Starting Cycle Dialog** SHALL present a **Confirm and Send** button and a Cancel action.

K. When the Study Coordinator confirms the **Select Starting Cycle Dialog**, the interface SHALL update the **Questionnaire Card** status to **Sent**.

L. If the Study Coordinator cancels the **Select Starting Cycle Dialog**, the **Questionnaire Card** SHALL remain unchanged.

Start Next Cycle

M. When a Study Coordinator selects **Start Next Cycle**, the interface SHALL update the **Questionnaire Card** status to **Sent** without displaying the **Select Starting Cycle Dialog**.

Delivery Failed Troubleshooting

N. When the **Questionnaire Status** is **Delivery Failed**, the interface SHALL display an information icon adjacent to the **Status Badge**. When the **Study Coordinator** selects this icon, the interface SHALL display a **Troubleshooting Popover** with guidance on resolving delivery issues.

O. The **Troubleshooting Popover** SHALL dismiss when the **Study Coordinator** clicks outside its bounds or selects a close action within it.

5.3.9 GUI-CAL-p00007: Questionnaire Finalization Workflow

Definitions:

Finalization Dialog: The Confirmation Dialog displayed to the Study Coordinator when the Finalize button is selected, containing a Cycle dropdown for selecting the Cycle value before confirming finalization.

Assertions:

Finalization Dialog

A. When a Study Coordinator selects **Finalize**, the interface SHALL display the **Finalization Dialog** containing a **Cycle** dropdown.

B. The **Cycle** dropdown SHALL present the following selectable options: the **Current Cycle N Day 1** value, **End of Treatment**, and **End of Study**.

C. The **Finalization Dialog** SHALL present a **Finalize Questionnaire** button and a Cancel action.

Finalization Outcomes

D. When the **Study Coordinator** finalizes the **Questionnaire** with a **Cycle N Day 1** value selected, the interface SHALL update the **Questionnaire Card** status to **Not Sent** and display the **Finalized Cycle** (paired with the finalization date and time) and the **Next Cycle** on the card.

E. When the Study Coordinator finalized the Questionnaire with a **Terminal Cycle** value (**End of Treatment** or **End of Study**) selected, the interface SHALL update the **Questionnaire Card** status to **Closed**, display the **Terminal Cycle** value alongside the **Closed** status as a combined badge.

F. When the Study Coordinator cancels the **Finalization Dialog**, the **Questionnaire** SHALL remain unchanged.

G. When the **Study Coordinator** cancels the **Terminal Cycle Warning Dialog**, the **Questionnaire** SHALL remain unchanged and the **Study Coordinator** SHALL be returned to the **Finalization Dialog**.

5.3.10 REQ-CAL-p00080: Questionnaire Cycle Tracking

Each questionnaire must be traceable to a specific protocol cycle for regulatory compliance and data integrity. Automatic cycle incrementing reduces manual error. Because the protocol allows only one questionnaire of each type per cycle, no two completed questionnaires of the same type for a given participant may share the same cycle value. Some participants start the study using paper records, meaning their first electronic questionnaire may not be Cycle 1 Day 1. A Study Coordinator may select the starting cycle when sending the first questionnaire of each type. Cycle tracking is independent per questionnaire type.

Definitions:

Cycle: The protocol event identifier assigned to a Questionnaire representing the treatment cycle during which it was administered. Valid values are Cycle N Day 1 (where N is a positive whole number), End of Treatment, and End of Study.

Current Cycle: The Cycle value assigned to the most recently sent Questionnaire of a given Questionnaire Type for a participant.

Finalized Cycle: The Cycle value of the most recently finalized Questionnaire of a given Questionnaire Type for a participant.

Next Cycle: The Cycle value that will be assigned when the next Questionnaire of that Questionnaire Type is sent. Always the next sequential Cycle N Day 1 value following the Finalized Cycle (the Finalized Cycle N incremented by 1).

Starting Cycle: The Cycle value assigned to the first Questionnaire of a given Questionnaire Type sent to a participant.

Terminal Cycle: A Cycle value of End of Treatment or End of Study indicating no further Questionnaires of that Questionnaire Type will be sent to the participant.

Cycle N Day 1: A Cycle value identifying the first day of the Nth treatment cycle, where N is a positive whole number (Cycle 1 Day 1, Cycle 2 Day 1, etc.).

Assertions:

Cycle Assignment

- A. The System SHALL assign a **Cycle** value to each **Questionnaire** at the time it is sent.
- B. The System SHALL NOT allow two **Questionnaires** of the same **Questionnaire Type** for the same participant to share the same **Cycle** value.
- C. When a Study Coordinator sends the first **Questionnaire** of a given **Questionnaire Type** to a participant, the System SHALL require the Study Coordinator to select a **Starting Cycle** before the **Questionnaire** is sent.
- D. For each subsequent **Questionnaire** of a given **Questionnaire Type**, the System SHALL assign the **Next Cycle** value automatically by incrementing the Cycle N value by 1.

E. When a **Questionnaire** is called back before finalization, the System SHALL reassign the same **Cycle** value to the next **Questionnaire** of that **Questionnaire Type** for that participant.

Terminal Cycles

F. When a **Questionnaire** is finalized with a **Terminal Cycle** value, the System SHALL change the **Questionnaire** status to **Closed**.

G. The System SHALL NOT allow a **Terminal Cycle** value to be assigned to more than one **Questionnaire** of the same **Questionnaire Type** per participant.

H. When the **Questionnaire** status is **Closed**, the System SHALL NOT permit a new **Questionnaire** of that **Questionnaire Type** to be sent to that participant.

Configuration

I. The System SHALL support per-sponsor configuration request of whether cycle tracking is enabled for a given deployment.

J. When cycle tracking is enabled, the System SHALL support per-sponsor configuration request of whether the Study Coordinator is required to select a **Starting Cycle** when sending the first **Questionnaire** of a given **Questionnaire Type** to a participant.

K. When the **Starting Cycle** selection is disabled, the System SHALL assign Cycle 1 Day 1 as the **Starting Cycle** for all participants.

5.3.11 REQ-CAL-p00085: Questionnaire Session Resume and Timeout

Session resume, expiry behavior, state preservation, and clock semantics are defined by REQ-p01073 (Questionnaire Session Timeout). This requirement specifies only the sponsor portal-specific configuration.

Assertions

- A. The **NOSE HHT** and **HHT-QoL** questionnaires SHALL be configured with a **Session Timeout** of 30 minutes.
- B. The **NOSE HHT** and **HHT-QoL** questionnaires SHALL be configured with a **Timeout Warning Notification** threshold of 5 minutes before expiry.

5.4 Participant Workflows

5.4.1 REQ-p70007: Linking Code Lifecycle Management

The Mobile Linking Code mechanism provides a secure, time-limited channel through which clinical staff can link a participant's device to the Sponsor Portal without requiring the participant to authenticate directly. Single-use enforcement and unpredictability ensure that codes cannot be reused or guessed, while configurable expiry balances security with the practical time participants need to complete the linking process.

Definitions

Mobile Linking Code: A unique, time-limited, single-use code generated by the System and provided to a Participant by clinical staff. The Participant enters the code in the Mobile Application to establish a connection between their device and the Sponsor Portal.

Participant Linking Code: The Mobile Linking Code that was used to establish the connection between a Participant's device and the Sponsor Portal. Displayed for reference and troubleshooting purposes. Cannot be used to establish a new connection.

Assertions

Generation

- A. When a Study Coordinator initiates the participant linking workflow, the System SHALL generate a unique **Mobile Linking Code** for that participant.
- B. The System SHALL generate each **Mobile Linking Code** such that it cannot be predicted or derived from any other known value.
- C. The System SHALL generate **Mobile Linking Codes** in a format that is short and alphanumeric to support easy communication to the participant.

Expiry

- D. The System SHALL expire a **Mobile Linking Code** after a configurable duration from the time of generation.
- E. When a new **Mobile Linking Code** is generated for a participant, the System SHALL immediately invalidate any previously active **Mobile Linking Code** for that participant.

Single-Use

- F. The System SHALL invalidate a **Mobile Linking Code** immediately upon successful use.

Rejection

- G. The System SHALL reject any attempt to link using a **Mobile Linking Code** that is expired, already used, or does not exist.

Visibility

- H. The System SHALL retain the **Participant Linking Code** against the Participant record after successful use for reference and troubleshooting purposes.

5.4.2 REQ-CAL-p70008: Linking Code Lifecycle Configuration

Assertions

A. The **Mobile Linking Code** expiry duration SHALL be configured to 72 hours from the time of generation.

5.4.3 REQ-p70013: Participant Registration

Clinical trial portals require a defined mechanism for making participant records available to clinical staff.

Definitions

Participant: An individual whose **Mobile Application** has been linked to the **Sponsor Portal** for the purpose of clinical trial data collection. A **Participant** has been registered in the **Sponsor Portal** as a clinical trial subject.

Assertions

- A. The System SHALL support a sponsor-configurable mechanism for adding **Participant** records to the **Sponsor Portal**.
- B. When a **Participant** record is added to the **Sponsor Portal**, the System SHALL assign an initial **Participant** status as defined by the sponsor configuration.

5.4.4 REQ-CAL-p00063: Participant Registration Configuration

The sponsor portal uses Rave EDC as the participant registration source. Near real-time synchronization ensures that participant records are available in the Sponsor Portal as soon as they are created in Rave EDC, without requiring manual intervention from clinical staff.

Definitions

Rave EDC: The electronic data capture system used by the portal as the authoritative source of participant and site records.

Participant ID: The unique identifier assigned to a Participant in **Rave EDC** and synchronized to the **Sponsor Portal**. The **Participant ID** is the primary reference used to identify a **Participant** across the System.

Assertions

- A. The System SHALL synchronize **Participant** records from **Rave EDC** to the **Sponsor Portal** in near real-time.
- B. The System SHALL synchronize the **Participant ID** and assigned **Site** for each **Participant** record.
- C. When a **Participant** record is synchronized from **Rave EDC** for the first time, the System SHALL assign an initial **Participant** status of **Not Connected**.
- D. The System SHALL NOT permit **Participant** records to be created, edited, or deleted manually in the **Sponsor Portal**.

5.4.5 REQ-p70009: Link New Participant Workflow

Linking connects a Participant record in the Sponsor Portal to the Participant's Mobile Application, enabling data synchronization and questionnaire distribution. The workflow is initiated by clinical staff and completed by the Participant entering the Mobile Linking Code in the Mobile Application.

Assertions

- A. The System SHALL allow a Study Coordinator to initiate the linking workflow for a **Participant** with **Not Connected** status.
- B. When a Study Coordinator initiates the linking workflow, the System SHALL generate a **Mobile Linking Code**.
- C. When a **Participant** successfully enters a valid **Mobile Linking Code** in the **Mobile Application** for the first time, the System SHALL link the **Participant's** device to the **Sponsor Portal**.

5.4.6 REQ-p70010: Participant Disconnection Workflow

Disconnection temporarily severs the link between a Participant's device and the Sponsor Portal without removing the Participant record or their historical data. Clinical staff may disconnect a Participant when there are issues with the device, technical problems, or other circumstances prevent normal trial participation.

Assertions

- A. The System SHALL allow a Study Coordinator to disconnect a **Participant** whose current status permits disconnection.
- B. The System SHALL require a reason before disconnection is applied.
- C. The System SHALL support sponsor-configurable reason format for disconnection, either as a free text field or a predefined list.
- D. The System SHALL preserve all **Participant** data and history upon disconnection.
- E. When a **Participant** is disconnected, the System SHALL stop synchronizing data from the **Participant's Mobile Application** to the **Sponsor Portal**.
- F. When a **Participant** is disconnected, sponsor-specific rules SHALL remain applied to the **Participant's Mobile Application**.

5.4.7 REQ-CAL-p00020: Disconnection Reason Options

Assertions

- A. The disconnection reason SHALL be selected from a predefined list.
- B. The predefined disconnection reason options SHALL be: Device Issues, Technical Issues, and Other.

5.4.8 REQ-p70011: Participant Reconnection Workflow

Reconnection restores a disconnected Participant's link to the Sponsor Portal, enabling them to resume trial participation. A new Mobile Linking Code is generated to ensure the previous invalidated code cannot be reused.

Assertions

- A. The System SHALL allow a Study Coordinator to reconnect a **Participant** with **Disconnected** status.
- B. The System SHALL require a reason before reconnection is applied.
- C. The reconnection reason SHALL be captured via a **Reason Dialog — Free Text** by default and is sponsor-configurable.
- D. When reconnection is confirmed, the System SHALL generate a new **Mobile Linking Code**.
- E. When a **Participant** successfully enters the new **Mobile Linking Code**, the System SHALL restore the **Participant's** link to the **Sponsor Portal**.
- F. When a **Participant's** link is restored, the System SHALL synchronize all data recorded on the **Participant's Mobile Application** during the disconnected period to the **Sponsor Portal**.

5.4.9 REQ-p70017: Mark as Not Participating

A Study Coordinator may mark a Participant as not participating when they complete or withdraw from the trial. This is distinct from disconnection — sponsor-specific rules are no longer applied to the Participant's Mobile Application and the Participant is not expected to resume participation unless explicitly reactivated.

Assertions

- A. The System SHALL allow a Study Coordinator to mark a **Participant** with **Disconnected** status as **Not Participating**.
- B. The System SHALL require a reason before the action is applied.
- C. The System SHALL support sponsor-configurable reason format for marking as not participating, either as a free text field or a predefined list.
- D. When the action is confirmed, the System SHALL stop applying sponsor-specific rules to the **Participant's Mobile Application**.

5.4.10 REQ-CAL-p00064: Mark as Not Participating Reason Options

Assertions

- A. The reason SHALL be selected from a predefined list.
- B. The predefined reason options SHALL be: Subject Withdrawal, Death, Protocol Treatment/Study Complete, and Other.

5.4.11 REQ-p70016: Reactivate Participant

A Study Coordinator may reactivate a Participant who was incorrectly marked as not participating or who has re-enrolled in the trial. Reactivation restores sponsor-specific rules to the Participant's Mobile Application and makes the standard reconnection workflow available.

Assertions

- A. The System SHALL allow a Study Coordinator to reactivate a **Participant** with **Not Participating** status.
- B. The System SHALL require a free text reason before the action is applied.
- C. When the action is confirmed, the System SHALL re-apply sponsor-specific rules to the **Participant's Mobile Application**.

5.4.12 GUI-p70001: Participant Dashboard

The Participant Dashboard is the primary surface for clinical staff to manage participant linking, trial status, and questionnaires. It consists of a participant list organised by tabs, a search input, and a Participant Actions Modal accessible from each participant row.

Definitions

Participant Actions Modal: A modal dialog displayed when a Study Coordinator selects a participant row, presenting the Participant ID and the secondary actions available for that participant's current status.

Assertions

Tab Organization

- A. The System SHALL organise participants into tabs based on their current **Participant** status.
- B. Each tab SHALL display a count of participants matching its status grouping.

Search

- C. The System SHALL provide a search input that filters participants by **Participant** ID.
- D. The search SHALL apply across all tabs regardless of the currently selected tab.
- E. The search SHALL support partial matching.
- F. The search SHALL update results in real time as the user types.
- G. When the search is active, each tab count SHALL update to reflect matching participants within that tab's status grouping.
- H. When no participants match the search, the **System** SHALL display the message "No results found."

Participant List

- I. The interface SHALL display the **Participant** ID, **Site**, current status as a **Status Badge**, and a primary action in the Action column for each participant row.
- J. The interface SHALL display the currently active tab with a visual indicator.
- K. When a **Participant's** status changes, the interface SHALL move the participant row to the appropriate tab immediately.

Participant Actions Modal

- L. The interface SHALL display the **Participant Actions Modal** when a Study Coordinator selects a participant row.
- M. The **Participant Actions Modal** SHALL display the **Participant** ID.
- N. The interface SHALL present the actions available in the **Participant Actions Modal** according to the action availability table.

Action Availability

O. The interface SHALL display the primary action in the Action column and the secondary actions in the **Participant Actions Modal** according to the action availability table defined in GUI-p03001.

5.4.13 GUI-p03001: Participant Dashboard Configuration

Definitions

Participant Summary: The participant list section of the Sponsor Portal dashboard displaying all participants across the Study Coordinator's assigned sites.

Assertions

Tabs

- A. The interface SHALL display three tabs: **Not Connected**, **Active**, and **Inactive**.
- B. The **Not Connected** tab SHALL display participants with status **Not Connected**, **Pending**, or **Expired**.
- C. The **Active** tab SHALL display participants with status **Linked - Awaiting Start** or **Trial Active**.
- D. The **Inactive** tab SHALL display participants with status **Disconnected** or **Not Participating**.
- E. The interface SHALL display the **Not Connected** tab by default.

Action Availability Table

F. The interface SHALL display the primary action in the Primary Action column and the secondary actions in the **Participant Actions Modal** according to the following table:

Status	Primary Action Column	Participant Actions Modal
Not Connected	Link Participant	Link Participant
Pending	Show Linking Code	Show Linking Code
Expired	Regenerate Code	Regenerate Code
Linked - Awaiting Start	Start Trial	Show Linking Code, Disconnect Participant
Trial Active	Manage Questionnaires	Show Linking Code, Disconnect Participant
Disconnected	Reconnect Participant	Reconnect Participant, Mark as Not Participating, Show Linking Code
Not Participating	Reactivate Participant	Reactivate Participant, Show Linking Code

Linking Code Display

G. When a Study Coordinator selects Show Linking Code for a Participant with Pending status, the interface SHALL display the **Mobile Linking Code** with a Copy action and a Save as PDF action.

H. The Save as PDF action SHALL generate a PDF containing the **Mobile Linking Code** and participant instructions.

I. When a Study Coordinator selects Show Linking Code for a Participant with any status other than Pending, the interface SHALL display the **Participant Linking Code**.

Link Participant Flow

J. When a Study Coordinator selects Link Participant, the interface SHALL display a Confirmation Dialog showing the **Participant** ID and the configured **Mobile Linking Code** expiry duration.

K. When the Study Coordinator confirms, the interface SHALL display an Acknowledgement Dialog showing the generated **Mobile Linking Code**, a Copy action, and the remaining time until expiry.

L. When the Study Coordinator dismisses the Acknowledgement Dialog, the interface SHALL update the **Participant's Status Badge** to **Pending**.

5.5 Sponsor Portal Authentication

5.5.1 REQ-p20068: Password Requirements

Definitions

Password: A secret string of characters used to authenticate a user's identity when accessing the **System**.

Password Reuse Limit: The configurable number of the user's most recent Passwords that a new Password must not match.

Assertions

- A. The **System** SHALL require a **Password** to be a minimum of 12 characters in length.
- B. The **System** SHALL require a **Password** to contain at least one uppercase letter, one lowercase letter, one numeric character, and one special character.
- C. The **System** SHALL reject a **Password** that is commonly used or easily guessable as per NIST SP 800-63B commonly used password list.
- D. The **System** SHALL require a **Password** to be changed after 90 days, unless a different interval has been configured for the deployment.
- E. The **System** SHALL prevent a user from accessing the **System** until their **Password** has been changed upon expiry.

5.5.2 REQ-p20069: Two-Factor Authentication

Two-Factor Authentication adds a second independent factor beyond the **Password** to prevent unauthorized access in the event of credential compromise. The platform supports configurable second factors so each sponsor deployment can select the method that best fits its operational and regulatory context.

Definitions

Second Factor: The independent authentication factor required in addition to the **Password** during login. The specific method is sponsor-configurable per deployment.

Verification Code: A single-use, time-limited code presented as the **Second Factor** during login.

Code Expiry: The configurable duration after which an unused **Verification Code** becomes invalid.

Assertions

Enforcement

- A. The **System** SHALL require Two-Factor Authentication for every login to the **Sponsor Portal**.
- B. The **System** SHALL NOT grant access to the **Sponsor Portal** until both the **Password** and a valid **Verification Code** have been successfully validated.

Verification Code Lifecycle

- C. When a **User Account** owner submits a valid **Password**, the **System** SHALL generate a **Verification Code** and deliver it via the configured **Second Factor** method.
- D. The **System** SHALL ensure each **Verification Code** is single-use and is invalidated immediately upon successful use.
- E. When a **Verification Code** has not been used within the **Code Expiry** duration, the **System** SHALL invalidate the **Verification Code**.
- F. When a **Verification Code** has been invalidated, the **System** SHALL require the **User Account** owner to restart the login process to receive a new **Verification Code**.

Configuration

- G. The **System** SHALL support sponsor-configurable selection of the **Second Factor** method per deployment.

H. The **System** SHALL support sponsor-configurable **Code Expiry** per deployment.

5.5.3 REQ-CAL-p00089: Two-Factor Authentication Configuration

Assertions

- A. The **Second Factor** method SHALL be configured to email delivery to the **User Account** owner's registered **Email Address**.
- B. The **Code Expiry** SHALL be configured to 10 minutes.

5.5.4 REQ-p20070: Forgot Password

A **User Account** owner who has forgotten their **Password** must be able to reset it without administrator intervention. The reset mechanism uses a time-limited, single-use **Verification Link** delivered to the registered **Email Address** to confirm the requester controls the account.

Assertions

Initiation

- A. The **System** SHALL allow a **User Account** owner to initiate a **Password** reset from the **Sponsor Portal** login interface by providing their **Email Address**.
- B. When a **Password** reset is initiated, the **System** SHALL generate a **Verification Link** and deliver it to the registered **Email Address**.
- C. The **System** SHALL display a confirmation that an email has been sent regardless of whether the **Email Address** matches an existing **User Account**.

Verification Link Behavior

- D. When a **Verification Link** issued for **Password** reset is used or 24 hours have passed from generation, whichever occurs first, the **System** SHALL invalidate the **Verification Link**.
- E. When a new **Password** reset **Verification Link** is issued for a **User Account**, the **System** SHALL invalidate any previously issued **Password** reset **Verification Link** for that **User Account**.

Reset Completion

- F. When a **User Account** owner accesses a valid **Verification Link**, the **System** SHALL allow them to set a new **Password**.
- G. Upon successful **Password** reset, the **System** SHALL terminate all active sessions associated with that **User Account**.
- H. The **System** SHALL require Two-Factor Authentication on the next login following a successful **Password** reset.

Rejection Behavior

- I. When a **User Account** owner attempts to access an invalidated **Verification Link**, the **System** SHALL reject the attempt and prompt the user to initiate a new **Password** reset.

5.5.5 GUI-p20072: Forgot Password Workflow Interface

The Forgot Password workflow spans four screens: the request screen where the **User Account** owner enters their **Email Address**, the confirmation screen shown after submission, the reset screen reached through a valid **Verification Link**, and the invalid link screen reached through an invalidated **Verification Link**. Consistent screen behavior across the flow ensures the **User Account** owner can complete recovery or recognize when to start over.

Assertions

Entry Point

A. The **Sponsor Portal** login interface SHALL present a Forgot Password action that navigates to the Forgot Password Request screen.

Forgot Password Request Screen

B. The Forgot Password Request screen SHALL present a single **Email Address** input field and a Submit action.

C. The Forgot Password Request screen SHALL present a Back to Login action that returns to the login interface.

D. The interface SHALL not enable the Submit action until a value has been entered in the **Email Address** field.

E. The interface SHALL validate that the entered value conforms to a valid email address format before submission is accepted.

F. When the Submit action is invoked, the interface SHALL navigate to the Confirmation screen.

Confirmation Screen

G. The Confirmation screen SHALL display a message indicating that an email has been sent if the submitted **Email Address** is associated with an account.

H. The Confirmation screen SHALL display the duration after which the **Verification Link** expires.

I. The Confirmation screen SHALL display guidance to check the spam folder if the email is not received.

J. The Confirmation screen SHALL present a Back to Login action that returns to the login interface.

K. The Confirmation screen SHALL display the same content regardless of whether the submitted **Email Address** matches an existing **User Account**.

Reset Screen

L. The Reset screen SHALL be reachable only by accessing a valid **Verification Link** delivered to the **User Account** owner's registered **Email Address**.

M. The Reset screen SHALL present a New **Password** field and a Confirm **Password** field.

N. Each **Password** field SHALL present a show/hide toggle that controls whether the entered value is displayed in plain text or masked.

O. The Reset screen SHALL present a Submit action.

P. The interface SHALL not enable the Submit action until both **Password** fields have been populated and contain matching values.

Q. When the Submit action is invoked and the **Password** is rejected for failing composition or reuse rules, the interface SHALL display an inline message identifying which rule was violated and SHALL not navigate away from the Reset screen.

R. When the Submit action is invoked and the **Password** is accepted, the interface SHALL navigate to the **Sponsor Portal** login interface and display a confirmation that the **Password** has been changed.

Invalid Link Screen

S. When a **User Account** owner accesses an invalidated **Verification Link**, the interface SHALL display the Invalid Link screen.

T. The Invalid Link screen SHALL display a message indicating that the **Verification Link** is no longer valid.

U. The Invalid Link screen SHALL present an action that navigates to the Forgot Password Request screen.

5.5.6 REQ-p20071: Session Management

The **Sponsor Portal** terminates inactive sessions to limit the window during which an unattended authenticated session could be exploited. Session limits also ensure that role and permission changes take effect within a bounded time.

Definitions

Session: An authenticated period during which a **User Account** owner can access the **Sponsor Portal** without re-entering credentials.

Session Idle Timeout: The configurable maximum duration of inactivity allowed during a **Session** before the **System** terminates it.

Assertions

Session Establishment

A. The **System** SHALL establish a **Session** when a **User Account** owner successfully completes Two-Factor Authentication.

Idle Timeout

B. The **System** SHALL track elapsed inactivity from the **User Account** owner's most recent interaction with the **Sponsor Portal**.

C. When the **Session Idle Timeout** is exceeded, the **System** SHALL terminate the **Session** and require re-authentication.

Termination

D. The **System** SHALL allow a **User Account** owner to explicitly terminate their **Session** by logging out.

E. When a **User Account** is deactivated, the **System** SHALL terminate all active **Sessions** associated with that **User Account** immediately.

F. When a **User Account's Role** or **Site** assignment is changed, the **System** SHALL terminate all active **Sessions** associated with that **User Account** immediately.

Configuration

G. The **System** SHALL support sponsor-configurable **Session Idle Timeout** per deployment, with a default of 10 minutes.

5.5.7 REQ-CAL-p00090: Login Attempt Rate Limiting Configuration

Assertions

- A. The **Login Rate Limit Threshold** SHALL be configured to 5 consecutive failed login attempts.
- B. The **Login Rate Limit Cooldown** SHALL be configured to 60 seconds.
- C. The Login Rate Limit error message SHALL display the text: "Too many failed attempts. Please wait 60 seconds before trying again."

5.6 Administrator Settings

This section defines the **Administrator Settings** surface, a read-only view available to Administrators that displays the current values of sponsor-configured platform parameters. The surface provides Administrators with visibility into how the deployment is configured without granting modification capability.

5.6.1 REQ-p20082: Administrator Settings Surface

The Sponsor Portal maintains a body of sponsor-configured platform parameters that govern System behavior. These parameters are defined in individual requirements throughout the specification and are typically static for the duration of a deployment. Administrators require visibility into the current values of these parameters to support troubleshooting, audit response, and operational verification, without needing to consult the specification or contact CureHHT personnel. The Administrator Settings surface consolidates these values into a single read-only view scoped to the Administrator role.

Definitions

Configuration Parameter: A named, sponsor-configured value that governs a specific System behavior, defined in a single source requirement and surfaced to Administrators for visibility.

Configuration Category: A grouping of related Configuration Parameters presented together within the Administrator Settings surface.

Administrator Settings: The read-only surface within the **Sponsor Portal** that displays Configuration Parameters to Administrators, organised by Configuration Category.

Assertions

Availability

A. The **System** SHALL provide an **Administrator Settings** surface within the **Sponsor Portal**.

Content

B. The **System** SHALL display each **Configuration Parameter** in **Administrator Settings** with its name, current value, and unit of measurement where applicable.

C. The **System** SHALL display each **Configuration Parameter** under exactly one **Configuration Category**.

D. The **System** SHALL display the current value of each **Configuration Parameter** as defined by the source requirement that establishes it.

Read-Only Behavior

E. The **System** SHALL NOT permit modification of any **Configuration Parameter** value from the **Administrator Settings** surface.

Configuration

F. The **System** SHALL support sponsor-configurable selection of which Configuration Parameters are surfaced in **Administrator Settings** per deployment.

G. The **System** SHALL support sponsor-configurable definition of Configuration Categories per deployment.

5.6.2 GUI-p20083: Administrator Settings Interface

The Administrator Settings surface is reached from the Settings action in the Administrator Dashboard header. The surface presents Configuration Parameters as a grouped list, with each Configuration Category as a section heading and each Configuration Parameter as a row beneath it. The interface communicates the read-only nature of the view so the Administrator does not expect to edit values directly.

Assertions

Display

- A. The interface SHALL display the **Administrator Settings** surface as a full-page view that replaces the **Administrator Dashboard** top-level tab content.
- B. The interface SHALL display a header on the **Administrator Settings** surface containing the title **Settings** and a back action that returns the **Administrator** to the previously active top-level tab.
- C. The interface SHALL display each **Configuration Category** as a labeled section.
- D. The interface SHALL display each **Configuration Parameter** as a row within its **Configuration Category** section, showing the parameter name, the current value, and the unit of measurement where applicable.
- E. The interface SHALL display **Configuration Parameters** in the order defined by the deployment configuration.

Read-Only Indication

- F. The interface SHALL display a notice on the **Administrator Settings** surface indicating that values are read-only and that changes require contacting CureHHT personnel.
- G. The interface SHALL NOT present any input control, edit action, or save action on the **Administrator Settings** surface.

5.6.3 REQ-CAL-p00094: Administrator Settings Configuration Inventory

This requirement defines the Configuration Parameters surfaced in **Administrator Settings** for the Sponsor Portal deployment, the Configuration Categories under which they are grouped, and the source requirement establishing each value.

Assertions

A. The **Administrator Settings** surface SHALL display the following Configuration Categories in the order listed: **Authentication and Sessions**, **Mobile Application Security**, **Diary Entry Rules**, **Questionnaire Sessions**, **Notifications and Reminders**, and **Rate Limiting**.

B. The **Administrator Settings** surface SHALL display the following Configuration Parameters under their respective Configuration Categories with the current values shown:

Authentication and Sessions

Parameter	Source Requirement
Sponsor Portal Session Idle Timeout	REQ-p20071
Password Expiry Interval	REQ-p20068
Two-Factor Authentication Code Expiry	REQ-CAL-p00089

Mobile Application Security

Parameter	Source Requirement
Mobile Linking Code Expiry	REQ-CAL-p70008
Application Lock Idle Timeout	REQ-CAL-p00092
Application Lock PIN Length	REQ-CAL-p00092
Application Lock Failed Attempt Threshold	REQ-CAL-p00092

Diary Entry Rules

Parameter	Source Requirement
Justification Threshold	REQ-CAL-p05001
Lock Threshold	REQ-CAL-p05001
Short Duration Threshold	REQ-CAL-p05003
Long Duration Threshold	REQ-CAL-p05003

Questionnaire Sessions

Parameter	Source Requirement
NOSE HHT and HHT-QoL Session Timeout	REQ-CAL-p00085
NOSE HHT and HHT-QoL Timeout Warning Threshold	REQ-CAL-p00085

Notifications and Reminders

Parameter	Source Requirement
Incomplete Record Lock Warning Offset	REQ-CAL-p00091
Yesterday Entry Reminder Time	REQ-CAL-p00093
Ongoing Epistaxis Event Reminder Interval	REQ-CAL-p00093

Rate Limiting

Parameter	Source Requirement
Login Rate Limit Threshold	REQ-CAL-p00090
Login Rate Limit Cooldown	REQ-CAL-p00090
Linking Code Rate Limit Threshold	REQ-CAL-p05011
Linking Code Rate Limit Cooldown	REQ-CAL-p05011

6. MOBILE APPLICATION REQUIREMENTS

6.1 Mobile Application Foundation

6.1.1 REQ-p00043: Diary Mobile Application

The mobile application serves dual purposes: personal health tracking for individual **Users** and compliant data capture for clinical trials. In personal use mode the application requires no account and stores data locally. In linked use mode the application synchronizes data with the sponsor portal for clinical trial participation.

Definitions

Mobile Application: The iOS and Android application provided by the System for participant-reported diary data capture and questionnaire completion. Operates in personal use mode without an account or in linked use mode connected to a **Sponsor Portal** deployment.

User: An individual using the **Mobile Application** in personal use mode, without an account and without a link to a clinical trial. A User who subsequently links to a study becomes a **Participant**.

Assertions

- A. The System SHALL provide a mobile application for iOS platforms, available via the iOS app store.
- B. The System SHALL provide a mobile application for Android platforms, available via the Android app store.
- C. The mobile application SHALL support full offline operation for core diary functions in both personal use mode and linked use mode.
- D. The mobile application SHALL NOT require account creation or login for personal use mode.
- E. The **User** SHALL retain control over locally-entered data, including the right to delete it from the device while in personal use mode.
- F. The mobile application SHALL obtain explicit **User** consent before synchronizing pre-existing local data to the **Sponsor Portal** upon linking.

6.1.2 REQ-p00006: Offline-First Data Entry

Offline-first architecture ensures reliable data collection regardless of network conditions. All **Users** benefit from local storage for immediate, reliable data entry. For **Participants** linked to clinical trials, automatic synchronization provides cloud backup and enables remote monitoring while maintaining the offline-first experience.

Assertions

- A. The System SHALL allow **Users** to create diary entries without requiring internet connectivity.
- B. The System SHALL allow **Users** to edit diary entries without requiring internet connectivity.
- C. The System SHALL allow **Users** to view their complete entry history without requiring internet connectivity.
- D. For linked **Participants**, the System SHALL indicate which entries have not yet synchronized to the **Sponsor Portal**.
- E. For linked **Participants**, the System SHALL automatically synchronize unsynchronized entries when network connectivity becomes available.

6.1.3 REQ-p01039: Diary Start Day Definition

Clinical trial **Participants** may need to record nosebleeds that occurred before their first Mobile Application usage. The **Diary Start Day** establishes the earliest date for which diary entries are valid, balancing the need for historical data capture with the requirement to maintain reliable data quality. The **Diary Start Day** is set automatically based on the **Participant's** actual entries, when a **Participant** records data for a date earlier than any prior entry, the **Diary Start Day** moves backward to that date. The **Diary Start Day** never moves forward, even if the earliest entry is subsequently deleted, to preserve the historical scope of the diary and ensure that documented gaps remain visible.

Definitions

Diary Start Day: The earliest date for which diary entries are valid for a given **User**.

Assertions

- A. The System SHALL establish and maintain a **Diary Start Day** for each **User**.
- B. The System SHALL set the **Diary Start Day** to the date of the earliest entry the **Participant** has ever recorded.
- C. When a **Participant** records an entry for a date earlier than the current **Diary Start Day**, the System SHALL move the **Diary Start Day** backward to that date.
- D. The System SHALL NOT allow the **Diary Start Day** to be set earlier than 365 days before mobile application installation on the current device.
- E. The **Diary Start Day** SHALL only move backward, never forward, once set.
- F. The System SHALL NOT prompt **Users** to set the **Diary Start Day** during onboarding.
- G. Data entered for dates before the trial start date SHALL NOT synchronize to the **Sponsor Portal** or Rave EDC.
- H. The **System** SHALL NOT allow the **Participant** to record an entry for a date in the future.
- I. The **System** SHALL display each calendar day between the **Diary Start Day** and the current day, exclusive of any day for which a **Daily Status** has been recorded, as a missing day in the Calendar.

6.1.4 REQ-p00045: Clinical Trial Privacy Policy

Definitions

Application Privacy Policy: The privacy policy governing the Mobile Application itself, covering platform-level data handling.

Clinical Trial Privacy Policy: The privacy policy governing a Participant's participation in a specific clinical trial, covering sponsor-side data handling. Sponsor-configurable per deployment.

The Mobile Application SHALL provide access to a **Clinical Trial Privacy Policy** governing sponsor-side data handling for a linked study, distinct from the **Application Privacy Policy**.

Assertions

- A. The System SHALL support a sponsor-configurable **Clinical Trial Privacy Policy** per deployment.
- B. The System SHALL make the **Clinical Trial Privacy Policy** accessible to the Participant only when the Participant is linked to a study or is in the process of linking to a study.
- C. The System SHALL retain the **Clinical Trial Privacy Policy** version that was in effect at the time of consent against the Participant record.

6.1.5 REQ-CAL-p00045: Clinical Trial Privacy Policy Configuration

Assertions

- A. The Clinical Trial Privacy Policy SHALL be made available at [URL placeholder].

6.1.6 REQ-p20079: Application Lock

The Mobile Application contains Participant-reported clinical data and identifying information that must be protected from unauthorized access on the Participant's device. An Application Lock requires the Participant to authenticate before each use, ensuring that someone who picks up an unattended device cannot view or modify diary entries. The lock is configurable so each sponsor deployment can select the authentication method appropriate to its risk profile and Participant population. When the configured method is unavailable on a Participant's device, a fallback to Device Authentication ensures continuity of access without compromising the protective intent.

Definitions

Application Lock: The state in which the **Mobile Application** requires the **Participant** to authenticate before any **Participant**-facing screen, action, or data is accessible.

PIN: A numeric secret of configurable length set by the **Participant** and used to release the **Application Lock**.

Device Authentication: Authentication performed by the device operating system using the credential the **Participant** has configured at the device level, including but not limited to device passcode, fingerprint, or face recognition.

Authentication Method: The mechanism required to release the **Application Lock**. The configured **Authentication Method** is one of: **PIN**, **Device Authentication**, or none.

Idle Timeout: The configurable elapsed time the **Mobile Application** may remain in the background before the **System** re-applies the **Application Lock** on return to the foreground.

Assertions

Lock Application

- A. The **System** SHALL apply the **Application Lock** when the **Mobile Application** is launched from a fully closed state.
- B. The **System** SHALL apply the **Application Lock** when the **Mobile Application** returns to the foreground after spending more than the **Idle Timeout** in the background.
- C. While the **Application Lock** is applied, the **System** SHALL NOT display **Participant** data and SHALL NOT permit any **Participant** action other than authentication.

Authentication Method

D. The **System** SHALL release the **Application Lock** only upon successful authentication using the configured **Authentication Method**.

E. When the configured **Authentication Method** is **PIN**, the **System** SHALL require the **Participant** to enter their **PIN** to release the **Application Lock**.

F. When the configured **Authentication Method** is **Device Authentication**, the **System** SHALL invoke the device operating system's authentication prompt to release the **Application Lock**.

G. When the configured **Authentication Method** is none, the **System** SHALL NOT apply the **Application Lock**.

PIN Setup and Reset

H. When the configured **Authentication Method** is **PIN** and the **Participant** has not yet set a **PIN**, the **System** SHALL require the **Participant** to set a **PIN** before any **Participant** data is accessible.

I. The **System** SHALL require the **Participant** to enter the **PIN** twice during setup and SHALL reject the setup when the two entries do not match.

J. The **System** SHALL allow the **Participant** to change their **PIN** from the **Mobile Application** settings, after first authenticating with their existing **PIN** or **Device Authentication**.

Failed Attempt Handling

K. The **System** SHALL track consecutive failed **PIN** entry attempts.

L. When the number of consecutive failed **PIN** entry attempts reaches the configured **Failed Attempt Threshold**, the **System** SHALL fall back to **Device Authentication**.

M. Upon successful **Device Authentication** following a fallback, the **System** SHALL require the **Participant** to reset their **PIN** before any **Participant** data is accessible.

N. The **System** SHALL reset the failed attempt counter upon successful authentication.

Configuration

O. The **System** SHALL support sponsor-configurable selection of the **Authentication Method** per deployment.

P. The **System** SHALL support sponsor-configurable **PIN** length per deployment.

Q. The **System** SHALL support sponsor-configurable **Idle Timeout** per deployment.

R. The **System** SHALL support sponsor-configurable **Failed Attempt Threshold** per deployment.

OPEN: Should the **Application Lock** be bypassed for **Push Notification** taps that lead to time-sensitive flows (e.g., Ongoing Epistaxis Event Reminder), or always enforced?

6.1.7 REQ-CAL-p00092: Application Lock Configuration

These values configure the platform mechanism defined in REQ-p20079 for the Callisto deployment.

Assertions

- A. The **Authentication Method** SHALL be configured to **PIN**.
- B. The **PIN** length SHALL be configured to 4 digits.
- C. The **Idle Timeout** SHALL be configured to 15 minutes.
- D. The **Failed Attempt Threshold** SHALL be configured to 5 consecutive failed attempts.

6.2 Questionnaires Overview

6.2.1 REQ-p01066: Daily Epistaxis Record Questionnaire

Daily epistaxis tracking is the core data collection activity in HHT clinical trials because nosebleed frequency, duration, and severity are the primary outcome measures for evaluating treatment efficacy. participant self-reporting captures events as they occur rather than relying on retrospective recall during clinical visits, supporting the Contemporaneous principle of ALCOA+. "No nosebleeds" and "Don't remember" entries are valid daily summaries that distinguish between confirmed absence of events and missing data, which is critical for data integrity in regulatory submissions.

Definitions:

Daily Epistaxis Record: The primary nosebleed tracking instrument used in the HHT Clinical Diary, derived from Clark et al. (2018) with modifications.

Epistaxis Event: A single nosebleed occurrence recorded by the participant with timing, duration, and intensity data.

Assertions:

- A. The system SHALL implement the **Daily Epistaxis Record** questionnaire for capturing participant-reported nosebleed events.
- B. The source instrument SHALL be documented with citation: Clark et al. "Nosebleeds in hereditary hemorrhagic telangiectasia: Development of a participant-completed daily eDiary." Laryngoscope Investig Otolaryngol. 2018. doi:10.1002/lio2.211.
- C. All modifications to the source instrument SHALL be documented and traceable to their justification.
- D. The instrument content, field definitions, and validation rules SHALL be implemented in a computer-readable data file that has been manually checked for accuracy.

6.2.2 REQ-p01067: NOSE HHT Questionnaire

The NOSE HHT (Nasal Outcome Score for Epistaxis in Hereditary Hemorrhagic Telangiectasia) is a validated 29-question participant-reported outcome measure assessing the physical, functional, and emotional impact of nosebleeds on HHT participants. The platform implements this instrument faithfully from the validated source. Questionnaire content is defined in the source document and transcribed once into a computer-readable data file — the data file is the single implementation reference, eliminating duplication and ensuring the application and the specification never diverge.

Definitions:

NOSE HHT: The Nasal Outcome Score for Epistaxis in Hereditary Hemorrhagic Telangiectasia, a validated clinical instrument published in JAMA Otolaryngology — Head and Neck Surgery, 2020.

True Copy: A complete, unaltered reproduction of the source document obtained directly from the original publisher URL.

Assertions:

- A. The system SHALL implement the **NOSE HHT** questionnaire as defined in the source document referenced in assertion B.
- B. A **True Copy** of the source document SHALL have been obtained from:
<https://jamanetwork.com/journals/jamaotolaryngology/fullarticle/2771847>
- C. The **True Copy** SHALL be referenced by the computer-readable data file that implements this questionnaire.
- D. The source document SHALL be transcribed into a computer-readable data file that has been manually checked for accuracy. This transcribed version will serve as the official reference for all questions, answer choices, and scoring rules used in the system.
- E. The System SHALL present the 29 **NOSE HHT** questions organized into the three categories defined in the source instrument: Physical, Functional, and Emotional.
- F. The Questionnaire Display Name for the NOSE HHT Questionnaire SHALL be "NOSE HHT Survey".

6.2.3 REQ-p01068: HHT-QoL Questionnaire

The HHT-QoL questionnaire is a validated 4-question participant-reported outcome measure assessing how nosebleeds and HHT-related problems affect participants' daily activities and social engagement. The platform implements this instrument faithfully from the validated source. Questionnaire content is defined in the source document and transcribed once into a computer-readable data file — the data file is the single implementation reference, eliminating duplication and ensuring the application and the specification never diverge.

Definitions:

HHT-QoL: A validated clinical instrument published in Blood Advances, 2022, measuring the impact of HHT symptoms on participants' daily and social activities.

Assertions:

- A. The system SHALL implement the **HHT-QoL** questionnaire as defined in the source document referenced in assertion B.
- B. A **True Copy** of the source document SHALL have been obtained from:
<https://ashpublications.org/bloodadvances/article/6/14/4301/484568>
- C. The **True Copy** SHALL be referenced by the computer-readable data file that implements this questionnaire.
- D. The source document SHALL be transcribed into a computer-readable data file that has been manually checked for accuracy. This transcribed version will serve as the official reference for all questions, answer choices, and scoring rules used in the system.
- E. The system SHALL emphasize the key phrases — interrupted, avoided, had to miss, and other than nosebleeds — in the **HHT-QoL** questions, matching the formatting of the original validated instrument.
- F. The Questionnaire Display Name for the HHT-QoL Questionnaire SHALL be "HHT Quality of Life Survey".

6.3 Participant Questionnaire Workflow

6.3.1 REQ-p02065: Portal-Sent Questionnaire Rules

Definitions:

Portal-Sent Questionnaire: A Questionnaire initiated by a Study Coordinator from the Sponsor Portal and delivered to the participant via push notification.

Preamble: The introductory content block presented to the participant before questionnaire questions begin informing the participant of the estimated completion time and that answers will not be saved until Submission.

Submission: The participant action that signals all questions are answered and the Questionnaire is ready for Study Coordinator review.

Finalization: The Study Coordinator action that permanently locks participant answers, triggers score calculation and pushes the data to Rave EDC.

Assertions:

Preamble

- A. The System SHALL present the **Preamble** to the participant each time the participant opens a **Portal-Sent Questionnaire**.
- B. The **Preamble** SHALL inform the participant of the estimated time required to complete the questionnaire.
- C. The **Preamble** SHALL inform the participant that their progress will be retained while completing the questionnaire and that answers are submitted only when they complete **Submission**.
- D. The System SHALL require the participant to confirm readiness before proceeding from the **Preamble** to the questions.
- E. When the participant indicates they are not ready, the System SHALL return the participant to the home screen.

Completion Rules

- F. The System SHALL present one question at a time during **Portal-Sent Questionnaire** completion.
- G. The System SHALL NOT permit the participant to skip any question in a **Portal-Sent Questionnaire**.
- H. The System SHALL preserve in-progress answers locally while the participant is completing the questionnaire and SHALL NOT commit answers as a Submission until the participant completes **Submission**.

I. The System SHALL make completion progress available to the participant throughout the questionnaire.

J. The System SHALL allow the participant to navigate back and forth between the questions before **Submission**.

K. The System SHALL allow the participant to exit the questionnaire at any point before **Submission** without losing in-progress answers, subject to the **Session Timeout** defined in REQ-p01073.

L. The System SHALL present a review of all answers to the participant before **Submission**, allowing the participant to modify any answer before proceeding.

Submission

M. The System SHALL require an explicit participant action - clicking the Submit button - to complete **Submission**.

Edit Rules

N. The System SHALL allow the participant to edit their answers with no time limit after **Submission** until **Finalization**.

O. The System SHALL NOT permit the participant to edit their answers after **Finalization**.

6.3.2 GUI-p00001: Portal-Sent Questionnaire Workflow

Definitions:

Review Screen: The screen presented to the participant after answering the final question, displaying all questions and selected answers before Submission.

Assertions:

Preamble

- A. The interface SHALL present the **Preamble** with two actions: an I'm Ready button and a Not Now button.
- B. When the participant selects Not Now, the interface SHALL navigate the participant to the home screen.
- C. When the participant selects I'm Ready, the interface SHALL navigate the participant to the first question.

Question Screen

- D. The interface SHALL present a progress indicator showing the participant's current position within the questionnaire (Question #X out of Y).
- E. The interface SHALL display the Questionnaire Display Name as the screen title on every question screen.
- F. The interface SHALL present a back navigation control on every question screen.
- G. The interface SHALL present a Next action on every question screen.
- H. The Next action SHALL be disabled until the participant selects an answer.
- I. When the participant selects an answer and selects Next, the interface SHALL advance to the next question.
- J. When the participant navigates back to a previously answered question, the interface SHALL display the previously selected answer.

Review Screen

- K. After the participant answers the final question, the interface SHALL present the **Review Screen**.
- L. The **Review Screen** SHALL present all questions and the participant's selected answers.
- M. The **Review Screen** SHALL allow the participant to navigate to any individual question to change their answer.
- N. The **Review Screen** SHALL present a submit action.

Editing an Answer

N. When the participant navigates to an individual question from the **Review Screen** and changes an answer, the interface SHALL present a Save button.

O. When the participant selects Save, the interface SHALL return the participant to the **Review Screen**.

Submission

P. When the participant confirms **Submission**, the interface SHALL display an Acknowledgement Dialog confirming the questionnaire has been submitted successfully.

Post-Submission Editing

Q. When the participant re-opens a submitted questionnaire before **Finalization**, the interface SHALL present the **Review Screen**.

Finalization

R. When the participant opens a finalized questionnaire, the interface SHALL present the questionnaire in a read-only state with no edit or submit actions available.

6.3.3 REQ-p01073: Questionnaire Session Timeout

A configurable session timeout allows each questionnaire to enforce appropriate completion constraints based on its length and clinical requirements.

Definitions:

Session Timeout: The configurable maximum duration of inactivity allowed during questionnaire completion. When exceeded, the session expires and the answers selected so far are not retained.

Session Expiry: The state reached when a participant has not completed a questionnaire within the configured Session Timeout duration.

Timeout Warning Notification: Push notification delivered to the participant when the Session Timeout is approaching.

Session Expiry Notification: Push notification delivered to the participant when Session Expiry has occurred.

Assertions:

Timeout Tracking

A. When a questionnaire is configured with a **Session Timeout**, the System SHALL track elapsed inactivity from the participant's most recent interaction with the questionnaire.

B. The System SHALL only advance the inactivity timer while the participant is not actively interacting with the questionnaire.

C. When the **Session Timeout** is exceeded before the participant submits the questionnaire, the System SHALL discard all answers selected so far.

Expiry Behavior

D. When a participant returns to a questionnaire in a state of **Session Expiry**, the System SHALL present the questionnaire from the beginning including the **Preamble**.

Notifications

E. When a questionnaire is configured with a **Session Timeout**, the System SHALL deliver a **Timeout Warning Notification** to the participant when the **Session Timeout** is approaching.

F. When a questionnaire is configured with a **Session Timeout**, the System SHALL deliver a **Session Expiry Notification** to the participant when **Session Expiry** has occurred.

State Preservation

G. When a questionnaire is not configured with a **Session Timeout**, the System SHALL preserve and restore the participant's in-progress answers on return with no timeout constraint.

H. When a questionnaire with a **Session Timeout** has not yet expired, the System SHALL return the participant to their in-progress questionnaire on return.

Configuration

I. Each questionnaire definition SHALL support optional configuration of **Session Timeout** duration.

J. Each questionnaire definition SHALL support configuration of the threshold before expiry at which the **Timeout Warning Notification** is delivered.

6.3.4 GUI-p00004: Questionnaire Session Expiry

Definitions:

Session Expiry Dialog: The Acknowledgement Dialog displayed to the participant when they open a questionnaire that has reached Session Expiry.

Assertions:

Timeout Warning

A. When a **Timeout Warning Notification** is delivered, the interface SHALL present it as a push notification indicating that the questionnaire session is approaching expiry.

Expired Session on Return

B. When the participant opens a questionnaire that has reached **Session Expiry**, the interface SHALL display a **Session Expiry Dialog** informing the participant that their session has expired and their previous answers were not saved.

C. The **Session Expiry Dialog** SHALL present a Start Again button and a Not Now button.

D. When the participant selects Start Again, the interface SHALL dismiss the **Session Expiry Dialog** and present the **Preamble**.

E. When the participant selects Not Now, the interface SHALL navigate the participant to the home screen.

Session Expiry Notification

F. When a **Session Expiry Notification** is delivered, the interface SHALL present it as a push notification indicating that the questionnaire session has expired.

Active Session on Return

G. When the participant returns to a questionnaire with a session that has not yet reached **Session Expiry**, the interface SHALL restore the participant to the question they were on when they left, with their in-progress answers intact.

6.3.5 GUI-CAL-p07002: Portal-Sent Questionnaire Call-Back — Participant Experience

Participants need clear, immediate feedback when a Portal-Sent Questionnaire is called back so they do not waste effort on an inactive questionnaire or become confused about its disappearance.

Definitions:

Call-Back Notice: An Acknowledgement Dialog informing the **Participant** that a **Portal-Sent Questionnaire** assigned to them has been called back by the study team.

Assertions:

Open Questionnaire Interruption

- A. When a **Portal-Sent Questionnaire** is called back while the **Participant** has it open, the interface SHALL display the Call-Back Notice.
- B. The Call-Back Notice SHALL NOT be dismissible by tapping outside the dialog.
- C. When the **Participant** acknowledges the Call-Back Notice, the interface SHALL navigate the **Participant** to the **Main Screen**.

Task List

- D. When a **Portal-Sent Questionnaire** has been called back, the interface SHALL remove the corresponding Questionnaire Task from the Task List.

Next Session Notification

- E. When a **Portal-Sent Questionnaire** has been called back and the **Participant** did not have it open at the time of call-back, the interface SHALL display the Call-Back Notice on the **Participant's** next Application session.

Offline Submission Attempt

- F. When the **Participant** attempts to submit a **Portal-Sent Questionnaire** that has been called back, the interface SHALL display the Call-Back Notice.

6.4 Daily eDiary: HHT Epistaxis Data Capture

6.4.1 REQ-p00042: HHT Epistaxis Data Capture Standard

The HHT Epistaxis Data Capture Standard defines the requirements for participant-reported nosebleed event collection in HHT clinical trials. Epistaxis frequency, duration, and severity are primary outcome measures for evaluating treatment efficacy. Timezone-aware timestamps ensure precise duration calculation regardless of participant location.

Definitions:

Daily Status: The participant's self-reported summary of nosebleed activity for a given day. Valid values are Had Nosebleed, No Nosebleed, and Don't Remember.

Intensity: A participant-reported measure of nosebleed severity captured on a defined six-level scale ordered from least to most severe.

Incomplete Record: An **Epistaxis Event** that has been saved but is missing one or more required fields: start time, **Intensity**, or end time.

Assertions:

- A. The System SHALL require a participant to record a **Daily Status** for each calendar day within the diary period.
- B. The System SHALL enforce mutual exclusivity of the three **Daily Status** values — a participant SHALL NOT record more than one **Daily Status** per diary entry.
- C. When a participant records a **Daily Status** of Had Nosebleed, the System SHALL require the participant to provide start time and end time for the **Epistaxis Event** with timezone information.
- D. The System SHALL calculate the duration of an **Epistaxis Event** from the recorded start time and end time, accounting for timezone differences.
- E. The System SHALL require the participant to record an **Intensity** value for each **Epistaxis Event**.
- F. The System SHALL present notes options from a predefined list (defined in REQ-CAL-p05001) for each **Epistaxis Event**; the available options SHALL be sponsor-configurable per study protocol.
- G. The System SHALL classify an **Epistaxis Event** as an **Incomplete Record** when it has been saved with one or more required fields not recorded.

6.4.2 GUI-p00002: Record Nosebleed Event

Definitions:

Progress Indicator: The persistent bar displayed at the top of every screen in the recording flow showing the three steps — Start, Max Intensity, and End — and their completion state.

Time Picker: The time selection component presenting the current time as default, a date selector, a timezone selector, and minute adjustment controls of -15, -5, -1, +1, +5, and +15 minutes.

Assertions:

Flow Structure

- A. The interface SHALL display the **Progress Indicator** on every screen throughout the recording flow showing the Start, Max Intensity, and End steps.
- B. The interface SHALL highlight the active step in the **Progress Indicator** and show the recorded value for each completed step.
- C. The interface SHALL allow the participant to navigate to any previously completed step by tapping it in the **Progress Indicator**.
- D. The interface SHALL present a Back navigation action on every screen throughout the recording flow.

Nosebleed Start Time

- E. The interface SHALL present a **Time Picker** for the participant to set the nosebleed start time.
- F. The interface SHALL present a Set Start Time button that confirms the start time and advances to the Max Intensity screen.

Max Intensity

- G. The interface SHALL present the six intensity options as a selectable grid: Spotting, Dripping, Dripping quickly, Steady stream, Pouring, and Gushing.
- H. The interface SHALL NOT advance to the End Time screen until the participant has selected one of the intensity options.
- I. When the participant selects an intensity option, the interface SHALL highlight the selected option and automatically advance to the End Time screen.

Nosebleed End Time

- J. The interface SHALL present a **Time Picker** for the participant to set the nosebleed end time.
- K. When the participant selects Set End Time, the interface SHALL save the record and return the participant to the home screen.

- L. The System SHALL validate that end time is greater than or equal to start time.
- M. The System SHALL restrict end time selection to times not exceeding the current real time.
- N. The System SHALL require confirmation when creating entries less than 2 minutes old.

6.4.3 GUI-p00003: Nosebleed Event Delete

Assertions:

Delete During Recording Flow

- A. The interface SHALL present a delete action on every screen throughout the recording flow.
- B. When the participant selects the delete action during the recording flow, the interface SHALL display a **Reason Dialog — Predefined List** before proceeding.
- C. When the participant confirms the delete reason, the interface SHALL discard the in-progress record and return the participant to the home screen.
- D. When the participant cancels the Reason Dialog, the interface SHALL return the participant to the screen they were on.

Delete From Event History

- E. When the participant selects the delete action on a saved **Epistaxis Event**, the interface SHALL display a **Reason Dialog — Predefined List** before proceeding.
- F. When the participant confirms the delete reason, the interface SHALL remove the event from the participant's diary and return the participant to the previous screen.
- G. When the participant cancels the Reason Dialog, the **Epistaxis Event** SHALL remain unchanged.

6.5 Score Calculation

6.4.1 REQ-p02075: Questionnaire Score Calculation

Validated clinical questionnaires require scoring according to published, protocol-defined algorithms to ensure data integrity and comparability across participants and studies. Storing scores with survey records ensures traceability and supports downstream Rave EDC synchronization and clinical analysis.

Assertions:

- A. The System SHALL calculate **Questionnaire** scores according to the validated algorithm defined for each **Questionnaire**.
- B. The System SHALL store the calculated score with the associated questionnaire record.
- C. The scoring algorithm SHALL be traceable to its source definition for each **Questionnaire Type**.

6.4.2 REQ-p02008: HHT-QoL Score Calculation

Assertions:

- A. The System SHALL calculate the **HHT-QoL** questionnaire score according to the algorithm defined in: Development and performance of a hereditary hemorrhagic telangiectasia-specific quality-of-life instrument, Blood Advances, 26 July 2022, Volume 6, Number 14, DOI 10.1182/bloodadvances.2022007748.

6.4.3 REQ-p02009: NOSE HHT Score Calculation

Assertions:

- A. The System SHALL calculate the **NOSE HHT** questionnaire score according to the algorithm defined in: Development and Validation of the Nasal Outcome Score for Epistaxis in Hereditary Hemorrhagic Telangiectasia (NOSE HHT), JAMA Otolaryngology – Head & Neck Surgery, November 1, 2020, Volume 146, Number 11, DOI 10.1001/jamaoto.2020.3040.

6.6 Diary Entry Rules

6.6.1 REQ-p05000: Time-Based Entry Restrictions

Clinical trial data integrity requires that participant-reported data be contemporaneous. As time passes after an event, recall accuracy degrades and the risk of unreliable data increases. A two-tier restriction model balances the need for late data capture (with documented justification) against the need to prevent entries that are too old to be clinically reliable. The justification tier preserves ALCOA+ compliance by requiring documented reasons for non-contemporaneous entries. The lock tier enforces a hard boundary beyond which no data modification is permitted.

Definitions

Justification Threshold: The configurable elapsed time from midnight (00:00) of the event date in the **Participant's** local timezone after which the **Participant** must provide a reason before creating or editing an entry for that date.

Lock Threshold: The configurable elapsed time from midnight (00:00) of the event date in the **Participant's** local timezone after which the date is fully locked and no creation, editing, or deletion of entries is permitted.

Entry Justification: A reason selected from a sponsor-defined predefined list that the **Participant** must provide when creating or editing an entry that has exceeded the **Justification Threshold**.

Assertions

Justification Tier

- A. When the elapsed time for an event date exceeds the **Justification Threshold** but has not exceeded the **Lock Threshold**, the System SHALL require the **Participant** to select an **Entry Justification** before saving a new entry or an edit to an existing entry for that date.
- B. The System SHALL NOT require an **Entry Justification** when the **Participant** deletes an entry within the justification window.
- C. The System SHALL NOT allow free-text entry for **Entry Justification**.
- D. The System SHALL store the selected **Entry Justification** with the entry record.

Lock Tier

- E. When the elapsed time for an event date exceeds the **Lock Threshold**, the System SHALL NOT permit creation of new entries for that date.
- F. When the elapsed time for an event date exceeds the **Lock Threshold**, the System SHALL NOT permit editing of existing entries on that date.
- G. When the elapsed time for an event date exceeds the **Lock Threshold**, the System SHALL NOT permit deletion of existing entries on that date.

Configuration

- H. The System SHALL support sponsor-configurable **Justification Threshold** per deployment.
- I. The System SHALL support sponsor-configurable **Lock Threshold** per deployment.

J. The **Lock Threshold** SHALL be greater than or equal to the **Justification Threshold**.

K. The System SHALL support sponsor-configurable definition of the **Entry Justification** predefined list per deployment.

L. When neither threshold is configured for a deployment, the System SHALL NOT apply time-based entry restrictions.

M. The **Lock Threshold** SHALL apply only to event dates on or after the trial start date.

6.6.2 REQ-CAL-p05001: Time-Based Entry Restrictions Configuration

The Callisto clinical protocol requires that nosebleed event entries older than one calendar day include documented justification for the timing discrepancy, and that entries older than 48 hours are locked entirely to maintain data reliability. These values configure the platform mechanism defined in REQ-p05000.

Assertions

- A. The **Justification Threshold** SHALL be configured to 24 hours.
- B. The **Lock Threshold** SHALL be configured to 48 hours.
- C. The **Entry Justification** predefined list SHALL contain exactly four options: "Entered from paper records," "Remembered specific event," "Estimated event," and "Other."
- D. The **Entry Justification** prompt SHALL display the text: "This is an event more than one day old. Please explain why you are adding/changing it now."

6.6.3 REQ-p05002: Duration Reasonableness Check

Unusually short or long event durations may indicate data entry errors. Confirmation prompts at configurable thresholds help ensure data accuracy while allowing legitimate outlier events to be recorded.

Definitions

Duration Reasonableness Check: A configurable validation rule that prompts the **Participant** to confirm the recorded duration of an event when it falls outside expected boundaries.

Short Duration Threshold: The configurable minimum duration below which a confirmation prompt is triggered.

Long Duration Threshold: The configurable maximum duration above which a confirmation prompt is triggered.

Assertions

Short Duration Check

- A. When the duration of an event is less than the **Short Duration Threshold**, the System SHALL prompt the **Participant** to confirm the duration before saving.
- B. The confirmation prompt SHALL provide a confirm option that saves the entry as entered and a reject option that returns the **Participant** to edit the duration.

Long Duration Check

- C. When the duration of an event is greater than the **Long Duration Threshold**, the System SHALL prompt the **Participant** to confirm the duration before saving.
- D. The confirmation prompt SHALL provide a confirm option that saves the entry as entered and a reject option that returns the **Participant** to edit the duration.

Prompt Timing

- E. The System SHALL display the confirmation prompt before the save operation completes.

Configuration

- F. The System SHALL support sponsor-configurable **Short Duration Threshold** per deployment.
- G. The System SHALL support sponsor-configurable **Long Duration Threshold** per deployment.
- H. The System SHALL support sponsor-configurable enablement of each check independently per deployment.
- I. When a check is not enabled for a deployment, the System SHALL NOT display the corresponding confirmation prompt.

6.6.4 REQ-CAL-p05003: Duration Reasonableness Check Configuration

The Callisto clinical protocol defines specific duration boundaries for nosebleed events to flag potential data entry errors. These values configure the platform mechanism defined in REQ-p05002.

Assertions

- A. The **Short Duration Threshold** SHALL be configured to 1 minute.
- B. The **Long Duration Threshold** SHALL be configured to 60 minutes.
- C. Both the short duration check and the long duration check SHALL be enabled.
- D. The short duration confirmation prompt SHALL display the text: "Duration is under 1 minute, is that correct?"
- E. The long duration confirmation prompt SHALL display the text: "Duration is over 1 hour, is that correct?"

6.6.5 REQ-p05008: Overlapping Event Detection and Resolution

Nosebleed events cannot physically overlap — a participant cannot have two simultaneous, independent nosebleeds. When a new record's time range intersects with an existing record, the system must detect the conflict and guide the participant through resolution before the record is committed. This ensures data integrity and prevents logically impossible entries from reaching the clinical dataset.

Definitions

Overlap: A state in which the time range of a new or edited entry intersects with the time range of one or more existing entries for the same **Participant**.

Conflicting Record: An existing entry whose time range intersects with the entry being created or edited.

Resolution: The **Participant's** choice of how to handle an **Overlap** between two entries.

Assertions

Detection

A. The System SHALL evaluate for potential **Overlap** when the **Participant** sets the start time of a new or edited entry, and SHALL confirm the **Overlap** when the **Participant** sets the end time.

B. The System SHALL evaluate **Overlap** by comparing the actual start and end timestamps of the new or edited entry against all existing entries for the same **Participant**.

C. The System SHALL evaluate **Overlap** against both completed entries and ongoing entries.

Pre-Resolution State

D. When an **Overlap** is detected, the System SHALL save the new entry to persistent storage immediately upon creation.

E. When an **Overlap** is detected, the System SHALL NOT display the new entry in the **Participant's** diary history until **Resolution** is complete.

Resolution Sequencing

F. The System SHALL present the **Participant** with one **Conflicting Record** at a time, starting with the oldest.

G. After each **Resolution** step, the System SHALL evaluate the resulting entry for further **Overlaps**.

H. The System SHALL repeat the **Resolution** flow until no **Overlaps** remain.

I. When all **Overlaps** are resolved, the System SHALL commit all resulting changes as a single transaction.

Resolution Options

J. For each **Overlap**, the System SHALL offer three resolution options: keep the new entry and discard the **Conflicting Record**, keep the **Conflicting Record** and discard the new entry, or merge both entries into one.

Merge Behavior

- K. When the **Participant** selects merge, the System SHALL construct the merged entry using the earliest start time and the latest end time across both entries.
- L. When either entry in a merge is ongoing, the merged entry SHALL be ongoing.
- M. When the **Participant** selects merge, the System SHALL assign the higher severity value from the two original entries to the merged entry.

Post-Resolution Confirmation

- N. After any resolution choice, the System SHALL present the resulting entry to the **Participant** for review and confirmation before committing.
- O. The **Participant** SHALL be able to edit the resulting entry before confirming.

6.6.6 GUI-p05009: Overlapping Event Resolution Flow

When an overlap is detected, the **Participant** needs early warning and a clear resolution path. Displaying a warning as soon as the start time indicates a potential conflict gives the **Participant** awareness without interrupting their recording flow. Once the full time range is established, the resolution screen provides a side-by-side comparison and straightforward actions to resolve the conflict.

Definitions

Resolution Screen: The screen presented to the **Participant** when an **Overlap** is confirmed, displaying the new entry and the **Conflicting Record** side by side with available resolution actions.

Assertions

Early Warning

- A. When a potential **Overlap** is detected after the **Participant** sets the start time, the interface SHALL display a warning indicating that overlapping events have been detected and the number of conflicting records.
- B. The warning SHALL NOT prevent the **Participant** from continuing the recording flow.

Resolution Trigger

- C. When the **Participant** sets the end time and the **Overlap** is confirmed, the interface SHALL navigate the **Participant** to the **Resolution Screen**.

Resolution Screen

- D. The interface SHALL present the **Resolution Screen** displaying the new entry and the **Conflicting Record** side by side.
- E. Each entry on the **Resolution Screen** SHALL display the time range, duration, and severity.
- F. When the **Conflicting Record** is older than the **Justification Threshold**, the System SHALL require an **Entry Justification** before saving the resolution.
- G. The **Resolution Screen** SHALL present a Keep New action and a Keep Existing action. The Resolution Screen SHALL present a Merge Records action only when the Conflicting Record's event date has not exceeded the **Lock Threshold**.
- H. The Merge Records action SHALL display a description indicating that both records will be combined into one event.

Post-Resolution

- I. When the Merge Records action is unavailable due to the **Lock Threshold**, the Resolution Screen SHALL display a message indicating that the conflicting record is locked and cannot be merged.
- J. After the **Participant** selects any resolution option (Keep New, Keep Existing, or Merge Records), the interface SHALL navigate to the standard **Record Nosebleed** screen pre-filled with the resulting entry data.

K. The **Record Nosebleed** screen SHALL allow the **Participant** to review and edit any field of the resulting entry before confirming.

Multiple Conflicts

L. When multiple **Overlaps** exist, the interface SHALL present one **Resolution Screen** at a time.

M. When all **Overlaps** are resolved and the **Participant** confirms the final entry, the interface SHALL return the **Participant** to the Main Screen.

6.7 Mobile Application Navigation and Screens

6.7.1 GUI-p05007: Main Screen Layout

The **Main Screen** is the **Participant's** primary interface for daily diary use. Organizing the screen into distinct zones ensures that urgent items (disconnection alerts and tasks) are always visible, diary content is scrollable, and the primary recording action is always accessible. When no tasks or alerts are active, the content area expands to use the full available space.

Definitions

Main Screen: The default screen displayed to the **Participant** upon opening the mobile application.

System Notice Area: A dedicated zone at the top of the Main Screen reserved for persistent, non-dismissable system notices that require the Participant's attention.

Assertions

Screen Zones

- A. The Main Screen SHALL display the following zones in order from top to bottom: System Notice Area (when applicable), top navigation bar, Task List (when tasks are active), content area, and fixed bottom actions.
- B. The System Notice Area SHALL appear above the top navigation bar.
- C. When the System Notice Area is displayed, it SHALL NOT reduce the space available for any other Main Screen zone.

Task List Area

- D. When the **Task List** contains more tasks than the available space allows, the **Task List** SHALL scroll within its allocated area.
- E. When no tasks are active and no **Disconnection Notification** is displayed, the content area SHALL expand to fill the available space.

Content Area

- F. The content area SHALL display diary entries grouped by date, showing Yesterday and Today sections.
- G. The content area SHALL be scrollable between the task area and the fixed bottom actions.
- H. When a date section contains no entries, the content area SHALL display a message indicating no events for that date.

Fixed Bottom Actions

- I. The **Main Screen** SHALL display a Record Nosebleed button fixed at the bottom, regardless of scroll position.
- J. The **Main Screen** SHALL display a Calendar button fixed at the bottom below the Record Nosebleed button, regardless of scroll position.

6.7.2 GUI-p05012: Mobile Application Navigation and Screens

The mobile application provides two menus accessible from the top navigation bar: a **User Menu** for account and study-related actions, and an **Application Menu** for application-level functions. Each menu leads to dedicated screens. The top navigation bar is visible on the **Main Screen** at all times.

Definitions

User Menu: The menu accessed from the right side of the top navigation bar, grouping account and study-related actions.

Application Menu: The menu accessed from the left side of the top navigation bar, grouping application-level functions.

Assertions

Top Navigation Bar

- A. The **Main Screen** SHALL display a top navigation bar containing the **Application Menu** access on the left, the sponsor or Application logo in the center, and the **User Menu** access on the right.
- B. The top navigation bar SHALL remain visible on the **Main Screen** at all times.
- C. Only one menu SHALL be open at a time. Opening one menu SHALL close the other if it is open.
- D. Tapping anywhere outside an open menu SHALL dismiss it.

User Menu

- E. The **User Menu** SHALL contain the following items: Join the Study, User Profile, and Help Center.
- F. When the **User** clicks the “Join the Study” button SHALL navigate to the linking code entry screen.
- G. When the **Participant** is linked to a study, “Join the Study” button SHALL not be visible on the menu.
- H. The Help Center screen SHALL display contact information for support.

User Profile Screen

- I. When the **User** is not linked to any study, the Clinical Trial section SHALL display a message indicating no active study link and guidance on how to join.
- J. When the **Participant** is linked to a study, the Clinical Trial section SHALL display the **Participation Status Badge**.

Application Menu

- K. The **Application Menu** SHALL contain the following items: Accessibility and Preferences, **Application Privacy Policy**, and Licenses.
- L. The **Application Privacy Policy** screen SHALL display or link to the **Application Privacy Policy**.
- M. The Licenses screen SHALL display open-source and third-party license information.

6.7.3 GUI-p05014: Calendar and Day View

The **Calendar** provides **Participants** with a monthly overview of their diary activity and a way to navigate to any past date to view, add, or edit entries. The **Day View** is the screen displayed when a **Participant** selects a date from the **Calendar**.

Definitions

Calendar: The monthly view modal displaying date states and allowing the **Participant** to navigate to a specific date.

Day View: The screen displayed when the **Participant** selects a date from the **Calendar**, showing the entries for that date or prompting the **Participant** to record a **Daily Status**.

Assertions

Calendar Modal

- A. When the **Participant** selects the Calendar button, the interface SHALL display the **Calendar** as a modal overlay.
- B. The **Calendar** SHALL display a monthly view with navigation arrows to move between months.
- C. Each date in the **Calendar** SHALL display a visual indicator representing its diary state.
- D. The **Calendar** SHALL display a legend identifying each visual indicator state.
- E. The **Calendar** SHALL display the following date states: nosebleed events recorded, no nosebleeds confirmed, unknown / don't remember, incomplete or missing data, not recorded, **locked**, and **today**.
- F. A date SHALL display the incomplete state when the **Participant** has one or more **Incomplete Records** for that date.
- G. A date SHALL display the missing state when the date is within the diary period (from Diary Start Day to yesterday) and the Participant has not recorded any Daily Status for that date.
- H. The **Calendar** SHALL present a close action that dismisses the modal and returns the **Participant** to the **Main Screen**.

Day View — No Daily Status Recorded

- I. When the selected date has no **Daily Status** recorded, the **Day View** SHALL display the date and the prompt "What happened on this day?" with three actions: Add nosebleed event, No nosebleed events, and I don't recall / unknown.
- J. When the **Participant** selects No nosebleed events, the interface SHALL record a **Daily Status** of No Nosebleed for that date.
- K. When the **Participant** selects I don't recall / unknown, the interface SHALL record a **Daily Status** of Don't Remember for that date.
- L. When the **Participant** selects Add nosebleed event, the interface SHALL navigate the **Participant** to the nosebleed recording flow with the date set to the selected date.

Day View — Daily Status Recorded

M. When the selected date has a **Daily Status** recorded, the **Day View** SHALL display the date, the total event count, an Add new event action, and a list of all entries for that date.

N. Each nosebleed entry in the list SHALL display the start time, timezone, severity, and duration.

O. Each entry in the list SHALL be selectable.

P. Submitted **Portal-Sent Questionnaires** SHALL appear in the entries list on the date they were submitted.

Entry Selection

Q. When the **Participant** selects an entry, the interface SHALL navigate to the nosebleed recording flow for editing if the entry has not exceeded the **Lock Threshold**, or display the entry details in a read-only state if it has.

Day View Navigation

R. The **Day View** SHALL present a back action that returns the **Participant** to the **Calendar**.

Locked Dates

S. When a date has exceeded the **Lock Threshold**, the **Calendar** SHALL display that date in a greyed-out visual style distinct from other date states.

T. When the **Participant** selects a locked date with no **Daily Status** recorded, the **Day View** SHALL display a message indicating no record exists for that date and that the date can no longer be edited. The three action buttons (Add nosebleed event, No nosebleed events, I don't recall / unknown) SHALL NOT be displayed.

U. When the **Participant** selects a locked date with a **Daily Status** recorded, the **Day View** SHALL display the existing entries in a read-only state and SHALL display a message indicating the date can no longer be edited. The Add new event action SHALL NOT be displayed.

6.8 Participant Tasks and Notifications

6.8.1 GUI-p05005: Participant Task List

The task list provides **Participants** with a prioritized set of actions requiring their attention. Displaying tasks prominently on the **Main Screen** and linking directly to the relevant flows reduces friction and encourages timely data entry. Tasks are automatically resolved as the **Participant** addresses them, keeping the list current.

Definitions

Task List: The prioritized set of actionable items displayed on the **Participant's Main Screen** requiring **Participant** attention.

Incomplete Records Task: A task indicating the **Participant** has one or more saved entries with missing required data.

Questionnaire Task: A task indicating a **Portal-Sent Questionnaire** has been sent to the **Participant** and has not yet been submitted.

Yesterday Reminder Task: A task prompting the **Participant** to record a **Daily Status** for the previous day.

Assertions

Task Types and Priority

A. The **Task List** SHALL support the following task types, displayed in the priority order listed:

Priority	Task Type	Trigger	Removal
1	Incomplete Records Task	Participant has one or more saved entries with missing required data	All incomplete entries are completed or deleted
2	Questionnaire Task	Study Coordinator sends a Portal-Sent Questionnaire	Sponsor finalizes the questionnaire
3	Yesterday Reminder Task	New day begins and the Participant has not recorded a Daily Status for the previous day	Participant records a Daily Status for the previous day

Incomplete Records Task

B. The **Incomplete Records Task** SHALL display the count of incomplete entries.

C. When the **Participant** selects the **Incomplete Records Task**, the interface SHALL navigate to a screen listing all incomplete entries.

D. From the incomplete entries screen, the **Participant** SHALL be able to complete or delete each entry.

E. The **Incomplete Records Task** SHALL persist regardless of the age of the incomplete entries.

Questionnaire Task

F. The interface SHALL display one **Questionnaire Task** per **Questionnaire Type**.

G. When the **Participant** selects a **Questionnaire Task**, the interface SHALL navigate to the questionnaire flow for that **Portal-Sent Questionnaire**.

H. After the Participant submits a **Portal-Sent Questionnaire**, the questionnaire SHALL be accessible as a record on the day it was submitted via the **Calendar**.

I. After the sponsor finalizes the **Portal-Sent Questionnaire**, the **Questionnaire Task** SHALL be removed from the **Task List**.

J. After **Submission**, the **Questionnaire Task** SHALL display a completed visual state indicating the questionnaire has been submitted and is awaiting sponsor review.

K. While the **Questionnaire Task** is in a submitted state, the Participant SHALL be able to select it to review and edit their answers.

Yesterday Reminder Task

J. The **Yesterday Reminder Task** SHALL present three response actions: Yes, No, and Don't Remember.

K. When the **Participant** selects No, the interface SHALL record a **Daily Status** of No Nosebleed for the previous day and remove the task.

L. When the **Participant** selects Don't Remember, the interface SHALL record a **Daily Status** of Don't Remember for the previous day and remove the task.

M. When the **Participant** selects Yes, the interface SHALL navigate the **Participant** to the nosebleed recording flow with the date set to the previous day.

N. The **Yesterday Reminder Task** SHALL NOT appear if the **Participant** has already recorded a **Daily Status** for the previous day.

General Behavior

O. Tasks SHALL be removed from the **Task List** when their removal condition is met.

P. The **Task List** SHALL update as conditions change without requiring the **Participant** to refresh.

6.8.2 REQ-p05004: Disconnection Notification

Participants need immediate, clear notification when their mobile Application is disconnected from the sponsor portal. A persistent, non-dismissible notification ensures the **Participant** is always aware of their disconnected state and cannot accidentally or intentionally hide it.

Definitions

Disconnection Notification: A persistent, non-dismissible notification displayed to the **Participant** when their mobile Application is disconnected from the **Sponsor Portal**.

Assertions

- A. When a **Participant's** status is **Disconnected**, the System SHALL display a **Disconnection Notification** on the **Main Screen**.
- B. The **Disconnection Notification** SHALL persist until the **Participant** is reconnected to the **Sponsor Portal**.
- C. The **Disconnection Notification** SHALL NOT be dismissible by the **Participant**.
- D. The System SHALL support sponsor-configurable message content for the **Disconnection Notification**.
- E. When no sponsor-specific message content is configured, the **Disconnection Notification** SHALL display the following default message: "Your connection with the study has been interrupted. Please contact your study site for assistance."

6.8.3 GUI-p00076: Participation Status Badge

The **Participation Status Badge** gives **Participants** a persistent, at-a-glance view of their clinical trial involvement. The badge adapts its display based on the **Participant's** current status so they always know whether they are connected, disconnected, or no longer participating in a study.

Definitions

Participation Status Badge: The visual component displayed in the Clinical Trial section of the user profile screen showing the **Participant's** current study participation state and associated details.

Assertions

Placement

- A. The interface SHALL display the **Participation Status Badge** in the Clinical Trial section of the user profile screen.

Linked State

- B. When the **Participant's** status is **Linked - Awaiting Start** or **Trial Active**, the **Participation Status Badge** SHALL display the sponsor logo, the **Participant's** linking code, and the date and time the **Participant** joined.
- C. The **Participation Status Badge** SHALL include a link to the **Clinical Trial Privacy Policy** from the moment the Participant links to the study, and the link SHALL remain available thereafter regardless of subsequent status changes.

Disconnected State

- D. When the **Participant's** status is **Disconnected**, the **Participation Status Badge** SHALL display a warning indicator, the **Participant's** current linking code, and a message that the connection has been interrupted.
- E. When the **Participant's** status is **Disconnected**, the **Participation Status Badge** SHALL present an Enter New Linking Code action that navigates the **Participant** to the linking code entry screen.

Not Participating State

- F. When the **Participant's** status is **Not Participating**, the **Participation Status Badge** SHALL display in an inactive visual style with the end date of participation.

Automatic Update

- G. The **Participation Status Badge** SHALL update automatically when the **Participant's** status changes.
- H. The System SHALL support sponsor-configurable display of the sponsor logo on the **Participation Status Badge** when the **Participant's** status is **Not Participating**.

6.8.4 REQ-p05015: Incomplete Record Lock Warning Notification

Once an **Incomplete Record** reaches the Lock Threshold defined in REQ-p05001, the Participant cannot complete, edit, or delete it — the record is permanently retained in its incomplete state. To give Participants a final opportunity to act before this irreversible lock, the **System** sends a push notification at a configurable time before the lock fires. This requirement defines the platform mechanism; the trigger offset and notification text are sponsor-configurable.

Definitions

Lock Warning Offset: The configurable elapsed time before the Lock Threshold at which the **System** sends a push notification to the Participant about an **Incomplete Record** that has not been completed or deleted.

Assertions

Notification Behavior

- A. The **System** SHALL send a push notification to the Participant when an **Incomplete Record** reaches the Lock Warning Offset before the Lock Threshold defined in REQ-p05001.
- B. The **System** SHALL send the notification only once per **Incomplete Record**.
- C. When a Participant completes or deletes an **Incomplete Record** before the Lock Warning Offset is reached, the **System** SHALL NOT send the notification for that record.

Configuration

- D. The **System** SHALL support a sponsor-configurable Lock Warning Offset per deployment.
- E. The **System** SHALL support sponsor-configurable notification text per deployment.
- F. When the Lock Warning Offset is not configured for a deployment, the **System** SHALL NOT send the notification.
- G. The Lock Warning Offset SHALL be less than the Lock Threshold defined in REQ-p05001.

6.8.5 REQ-CAL-p00091: Incomplete Record Lock Warning Notification Configuration

The clinical protocol defines a 48-hour lock on diary entries (REQ-CAL-p05001). To give Participants a final opportunity to complete or delete an **Incomplete Record** before the lock fires, a warning notification is 24 hours prior. These values configure the platform mechanism defined in REQ-p05015.

Assertions

- A. The Lock Warning Offset SHALL be configured to 24 hours.
- B. The notification text SHALL be: "You have an incomplete nosebleed record from [date]. Complete or delete it within the next 24 hours, otherwise it will be permanently locked and you will not be able to change or remove it."

6.8.6 REQ-p05018: Portal-Sent Questionnaire Notification

When a **Study Coordinator** sends a **Portal-Sent Questionnaire** from the Sponsor Portal, the **Participant** must be made aware that a new questionnaire is available for completion. A push notification serves as the awareness mechanism. The Questionnaire Task on the **Main Screen** provides the **Participant**'s entry point into the questionnaire flow once the application is open.

Assertions

Trigger

A. When a **Portal-Sent Questionnaire** is successfully delivered to the **Mobile Application**, the System SHALL deliver a **Push Notification** to the **Participant**.

Offline Delivery

B. When the **Mobile Application** is offline at the time a **Portal-Sent Questionnaire** is sent, the System SHALL deliver the **Push Notification** when the **Mobile Application** next establishes connectivity.

Suppression

C. The System SHALL NOT deliver a **Push Notification** for a **Portal-Sent Questionnaire** that has already been submitted by the **Participant**.

D. The System SHALL NOT deliver a **Push Notification** for a **Portal-Sent Questionnaire** that has been called back by the **Study Coordinator**.

6.8.7 REQ-p05016: Yesterday Entry Reminder Notification

Clinical trial data completeness under ALCOA+ principles requires a **Daily Status** for each calendar day in the diary period. A daily reminder fired at a configured time of day prompts the **Participant** to review the previous day and respond via the Yesterday Reminder Task on the **Main Screen**.

Definitions

Reminder Time: The configurable time of day, in the **Participant's** local timezone, at which the Yesterday Entry Reminder Notification is delivered.

Assertions

Trigger

A. The System SHALL deliver a **Push Notification** at the **Reminder Time** when no **Daily Status** has been recorded for the previous calendar day.

Timezone

B. The System SHALL evaluate the **Reminder Time** against the **Participant's** device local timezone.

C. The System SHALL evaluate "the previous calendar day" against the **Participant's** device local timezone.

Suppression

D. The System SHALL NOT deliver the **Push Notification** when a **Daily Status** has been recorded for the previous calendar day.

E. The System SHALL deliver at most one Yesterday Entry Reminder Notification per calendar day.

Configuration

F. The System SHALL support sponsor-configurable **Reminder Time** per deployment.

6.8.8 REQ-p05017: Ongoing Epistaxis Event Reminder

A **Participant** who starts recording an **Epistaxis Event** but does not complete the record may forget to return and capture the end time, resulting in inaccurate duration data. A configured sequence of escalating reminders prompts the **Participant** to complete the record or confirm it is still ongoing. Once the sequence has fired, no further reminders are sent.

The **Participant** may configure their own reminder schedule for personal use. A sponsor-configured schedule overrides the personal setting for the duration of a clinical trial.

Definitions

Reminder Schedule: An ordered list of elapsed-time intervals defining the timing of successive reminder notifications for an Incomplete Record. Each interval is measured from the time of the previous notification, with the first interval measured from the Participant's most recent interaction with the record.

Assertions

Trigger and Sequence

- A. The **System** SHALL track the elapsed time since the **Participant's** most recent interaction with each **Incomplete Record**.
- B. The **System** SHALL deliver a **Push Notification** to the **Participant** at each interval in the **Reminder Schedule**.
- C. After the final interval in the **Reminder Schedule** has elapsed, the **System** SHALL NOT deliver further reminders for that **Incomplete Record**.

Reset on Interaction

- D. When the **Participant** interacts with an **Incomplete Record**, the **System** SHALL restart the **Reminder Schedule** from the first interval.

Termination

- E. When an **Incomplete Record** is completed, the **System** SHALL stop delivering reminders for that record.
- F. When an **Incomplete Record** is deleted, the **System** SHALL stop delivering reminders for that record.

Configuration

- G. The default **Reminder Schedule** for the **Mobile Application** SHALL be empty, resulting in no reminders being delivered.
- H. The **System** SHALL allow a **Participant** to configure their own **Reminder Schedule** in **Mobile Application** settings.
- I. The **System** SHALL support sponsor-configurable **Reminder Schedule** per deployment.
- J. When a sponsor-configured **Reminder Schedule** is in effect, the **System** SHALL apply the sponsor-configured schedule and SHALL NOT apply the **Participant's** personally configured schedule.

6.8.9 REQ-CAL-p00093: Notification & Reminder Configuration

Assertions

- A. The **Reminder Time** for the Yesterday Entry Reminder Notification SHALL be configured to 09:00 (24H format) in the **Participant's** local timezone.
- B. The **Reminder Schedule** for the Ongoing Epistaxis Event Reminder SHALL be configured to deliver notifications at the following intervals: 5 minutes, 10 minutes, 15 minutes, 30 minutes.

6.8.10 REQ-p05019: Historical Gap Reminder

The Historical Gap Reminder addresses missing **Daily Status** entries on days older than yesterday but still within the editable window.

Definitions

Historical Gap: A calendar day between the **Diary Start Day** and the current day, exclusive of the current day and the previous day, for which the **Participant** has not recorded a **Daily Status**.

Historical Gap Reminder: A daily **Push Notification** delivered to the **Participant** prompting them to address one or more **Historical Gaps**.

Assertions

Trigger

- A. The **System** SHALL evaluate, once per calendar day at the configured **Reminder Time**, whether the **Participant** has any **Historical Gap** within the editable window.
- B. When the **Participant** has at least one **Historical Gap** within the editable window, the **System** SHALL deliver a **Historical Gap Reminder** to the **Participant**.
- C. The **System** SHALL deliver at most one **Historical Gap Reminder** per calendar day.

Editable Window

- D. The **System** SHALL exclude from the **Historical Gap** evaluation any day for which the elapsed time has exceeded the **Lock Threshold**.
- E. In linked use mode, the **System** SHALL exclude from the **Historical Gap** evaluation any day before the trial start date.

Mode-Dependent Default

- E. In personal use mode, the **Historical Gap Reminder** SHALL be disabled by default.
- F. In personal use mode, the **System** SHALL allow the **User** to enable or disable the **Historical Gap Reminder** from the Mobile Application settings.
- G. In linked use mode, the **Historical Gap Reminder** SHALL be enabled by default.

Configuration

- H. The **System** SHALL support sponsor-configurable **Reminder Time** per deployment.

I. The **System** SHALL support sponsor-configurable notification text per deployment.

6.8.11 REQ-CAL-p00093: Historical Gap Reminder Configuration

Assertions

- A. The **Reminder Time** for the Historical Gap Reminder SHALL be configured to 09:00 (24H format) in the **Participant's** local timezone.
- B. The **Historical Gap Reminder** notification text SHALL be: "You have one day without a recorded entry. Tap to review and complete your diary."

6.9 Device Linking

6.9.1 REQ-p05010: Linking Code Entry Error Handling

Security best practices require generic error messages that do not reveal whether a linking code exists, is expired, or has been used. This prevents malicious actors from enumerating valid codes or determining code status. Rate limiting protects against brute-force attacks while a configurable cooldown balances security with usability.

Definitions

Linking Code Validation Failure: Any unsuccessful attempt to enter a linking code, regardless of the specific reason for failure.

Rate Limit Threshold: The configurable maximum number of consecutive **Linking Code Validation Failures** permitted within a single rate limit window before further attempts are temporarily blocked.

Rate Limit Cooldown: The configurable duration after which the rate limit counter resets and the **User** may attempt linking again.

Assertions

Error Messaging

- A. When a **Linking Code Validation Failure** occurs, the System SHALL display a single generic error message regardless of the specific reason for failure.
- B. The error message SHALL NOT reveal the specific reason for failure.
- C. When a **Linking Code Validation Failure** occurs, the System SHALL provide guidance to contact the site for a new code.

Rate Limiting

- D. When the number of consecutive **Linking Code Validation Failures** reaches the **Rate Limit Threshold**, the System SHALL block further linking attempts and display the **Rate Limit** error message until the **Rate Limit Cooldown** has elapsed.
- E. The **Rate Limit Cooldown** SHALL be calculated from the first failed attempt in the current window.

Configuration

- F. The System SHALL support sponsor-configurable **Rate Limit Threshold** per deployment.
- G. The System SHALL support sponsor-configurable **Rate Limit Cooldown** per deployment.
- H. The System SHALL support sponsor-configurable error message text per deployment.

6.9.2 REQ-CAL-p05011: Linking Code Entry Error Handling Configuration

These values configure the platform linking code error handling mechanism defined in REQ-p05010.

Assertions

- A. The **Rate Limit Threshold** SHALL be configured to 10 attempts.
- B. The **Rate Limit Cooldown** SHALL be configured to 1 hour.
- C. The **Linking Code Validation Failure** error message SHALL display the text: "Invalid linking code."
- D. The **Rate Limit** error message SHALL display the text: "Too many attempts. Please wait an hour."

6.9.3 GUI-p05013: Join the Study Screen

The **Join the Study** screen is presented to the User when initiating the linking workflow. It captures the **Mobile Linking Code** and the Participant's explicit consent to the **Clinical Trial Privacy Policy** before any link is established.

Definitions

Linking Consent: The explicit acknowledgement by the Participant that they have read and consent to the Clinical Trial Privacy Policy. This consent must be captured before the linking code submission is permitted.

Assertions

- A. The **Join the Study** screen SHALL display a **Mobile Linking Code** entry field formatted according to the configured code format.
- B. The screen SHALL display a **Linking Consent** checkbox with text confirming the Participant has read, understood, and consents to the **Clinical Trial Privacy Policy**.
- C. The **Linking Consent** text SHALL include a link that opens the **Clinical Trial Privacy Policy**.
- D. The **Submit** action SHALL be disabled until both a complete **Mobile Linking Code** is entered and the **Linking Consent** checkbox is checked.
- E. The System SHALL retain the **Linking Consent** acknowledgement, including the **Clinical Trial Privacy Policy** version, against the **Participant** record upon successful link.

6.9.4 GUI-p05015: Successful Linking Confirmation

When a Participant successfully enters a valid Mobile Linking Code on the Join the Study screen, the interface needs to clearly confirm that the link was established and transition the Participant from personal use mode into linked use mode.

Assertions

- A. When the **Participant** successfully submits a valid **Mobile Linking Code**, the interface SHALL display an **Acknowledgement Dialog** confirming that the device has been linked to the study.
- B. When the **Participant** acknowledges the dialog, the interface SHALL navigate the **Participant** to the **User Profile** screen.

7. APPENDICES

7.1 Screens

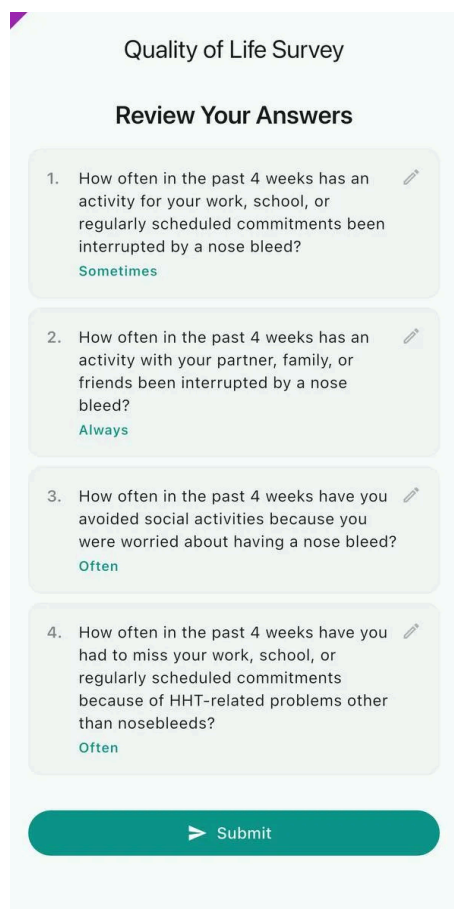
7.1.1 Questionnaire Review Screen

Trigger: **Participant** answers the final question of a **Portal-Sent Questionnaire**, before **Submission**.

Displays: **Questionnaire Display Name**, every question in original order, the **Participant's** selected response beneath each question, an Edit affordance per question, and a **Submit** action.

Outcome: Selecting **Submit** completes **Submission**.

Reference: GUI-p00001.



Quality of Life Survey

Review Your Answers

- How often in the past 4 weeks has an activity for your work, school, or regularly scheduled commitments been interrupted by a nose bleed?
Sometimes
- How often in the past 4 weeks has an activity with your partner, family, or friends been interrupted by a nose bleed?
Always
- How often in the past 4 weeks have you avoided social activities because you were worried about having a nose bleed?
Often
- How often in the past 4 weeks have you had to miss your work, school, or regularly scheduled commitments because of HHT-related problems other than nosebleeds?
Often

Submit

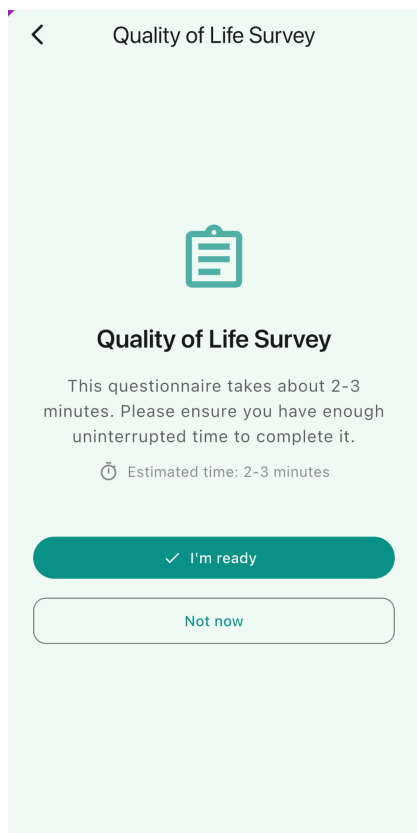
7.1.2 Questionnaire Preamble Screen

Trigger: Participant opens a **Portal-Sent Questionnaire**.

Displays: **Questionnaire Display Name**, estimated time to complete.

Actions: **I'm Ready** (proceed to first question), **Not Now** (return to **Main Screen**).

Reference: REQ-p02065, GUI-p00001.



< Quality of Life Survey



Quality of Life Survey

This questionnaire takes about 2-3 minutes. Please ensure you have enough uninterrupted time to complete it.

 Estimated time: 2-3 minutes

✓ I'm ready

Not now

7.1.3 Resolve Conflict — Resolution Screen

Trigger: New or edited entry **Overlaps** with one or more **Conflicting Records**. **Displays:** New entry and **Conflicting Record** side by side, showing time range, duration, and severity for each.

Actions: **Keep New** (retain new entry, discard **Conflicting Record**), **Keep Existing** (retain **Conflicting Record**, discard new entry), **Merge Records** (combine both using earliest start time, latest end time, and higher severity). **Conditional behavior:**

- **Merge Records** is presented only when the **Conflicting Record's** event date has not exceeded the **Lock Threshold**.
- When the **Conflicting Record** is older than the **Justification Threshold**, an **Entry Justification** is required before saving the resolution.

Outcome: **Participant** is navigated to the **Record Nosebleed** screen pre-filled with the resulting entry data for review and confirmation.

Reference: REQ-p05008, GUI-p05009.

Resolve Conflict

Two records overlap. Choose how to handle them.

New Record	Existing Record
Time 12:28 PM - 12:43 PM Duration 15m Severity  Dripping quickly	Time 12:27 PM - 12:43 PM Duration 15m Severity  Dripping quickly
Keep New	Keep Existing
 Merge Records Combine both into one event with the longest duration	

7.1.4 Troubleshooting Popover

Trigger: **Study Coordinator** selects the information icon adjacent to the **Status Badge** on a **Questionnaire Card** in **Delivery Failed** status.

Displays: Guidance for resolving delivery issues for the affected **Questionnaire**.

Dismissal: Click outside the popover.

Reference: GUI-CAL-p00006.

Immediate Troubleshooting Steps

1. Check the patient's phone is online — confirm Wi-Fi or cellular connection is active.
2. Check for a captive portal — if on hospital/hotel Wi-Fi, the phone may appear "connected" but require logging in through a browser page (accepting terms, entering a code, etc.). Open a browser and complete any login page.
3. Wait ~1 minute after connectivity is restored — the system will automatically retry delivery.
4. Check the app itself — confirm the app launches, isn't crashing, and the phone isn't out of storage (the app should surface these itself, but worth verifying).
5. If still failing after the above — fall back to a paper form and contact support / the study team per escalation protocol.

7.2 Modals

7.2.1 User Information Modal

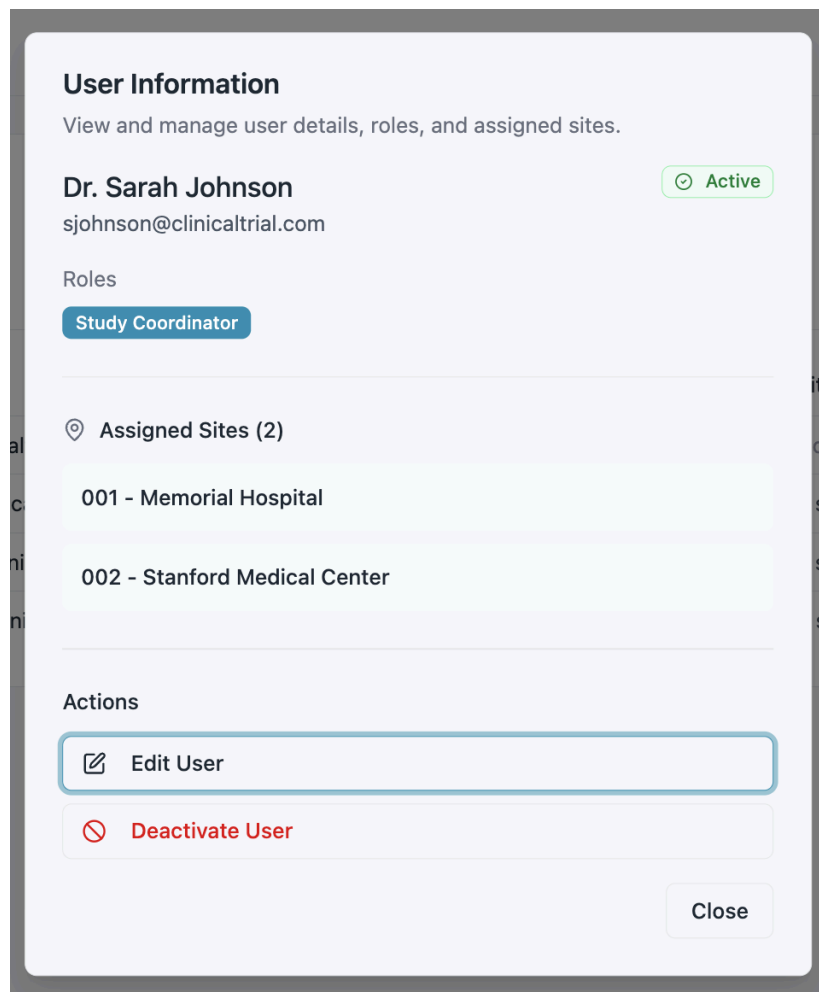
Trigger: **Administrator** selects a **User Account** row from the User Management interface.

Target identification: **Full Name, Email Address, Status.**

Displays: Assigned **Role(s)** and **Site** assignments. When the **User Account** holds one or more non-**Administrator Roles**, the **Administrator** may select a single **Role** to scope the displayed **Site** list. When the **User Account** is an **Administrator**, the modal indicates access to all **Sites**. *Actions (Active Users tab):* **Edit User, Deactivate User, Close.** The **Deactivate User** action is not presented for the currently authenticated user's own account.

Actions (Inactive Users tab): **Reactivate User, Close.**

Reference: GUI-p00033.



The screenshot shows a modal titled "User Information" with the subtitle "View and manage user details, roles, and assigned sites." The user profile for "Dr. Sarah Johnson" is displayed, with the email "sjohnson@clinicaltrial.com" and a green "Active" status badge. Under the "Roles" section, a blue pill-shaped button labeled "Study Coordinator" is shown. The "Assigned Sites (2)" section lists two sites: "001 - Memorial Hospital" and "002 - Stanford Medical Center". The "Actions" section contains two buttons: "Edit User" (with a pencil icon) and "Deactivate User" (with a red prohibition icon). A "Close" button is located at the bottom right of the modal.

7.2.2 Show Linking Code

Trigger: **Study Coordinator** selects **Show Linking Code** for a **Participant**.

Target identification: **Participant ID**.

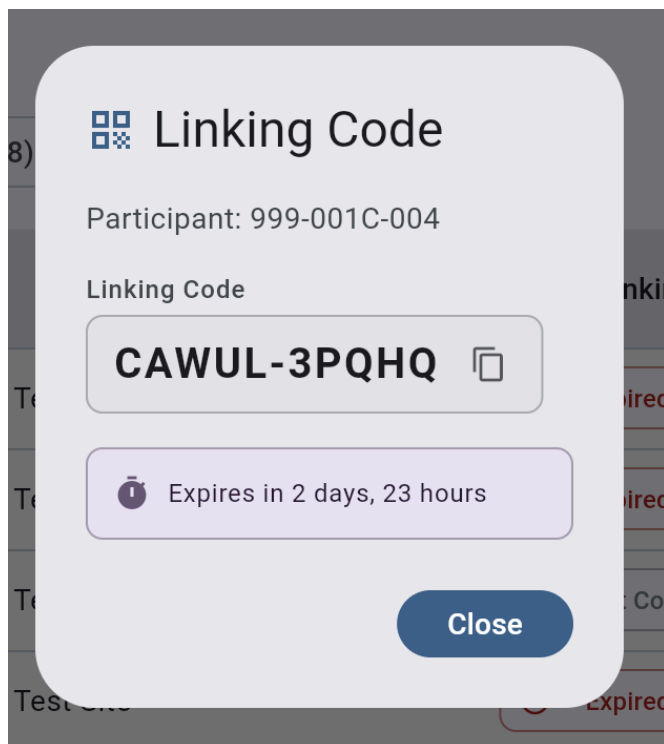
Variants:

- **Pending** status: displays the active **Mobile Linking Code** with **Copy** and **Save as PDF** actions. **Save as PDF** generates a PDF containing the **Mobile Linking Code** and **Participant** instructions.
- Any other status: displays the **Participant Linking Code** (the code previously used to establish the connection) with a **Copy** action, and indicates the code is shown for reference only.

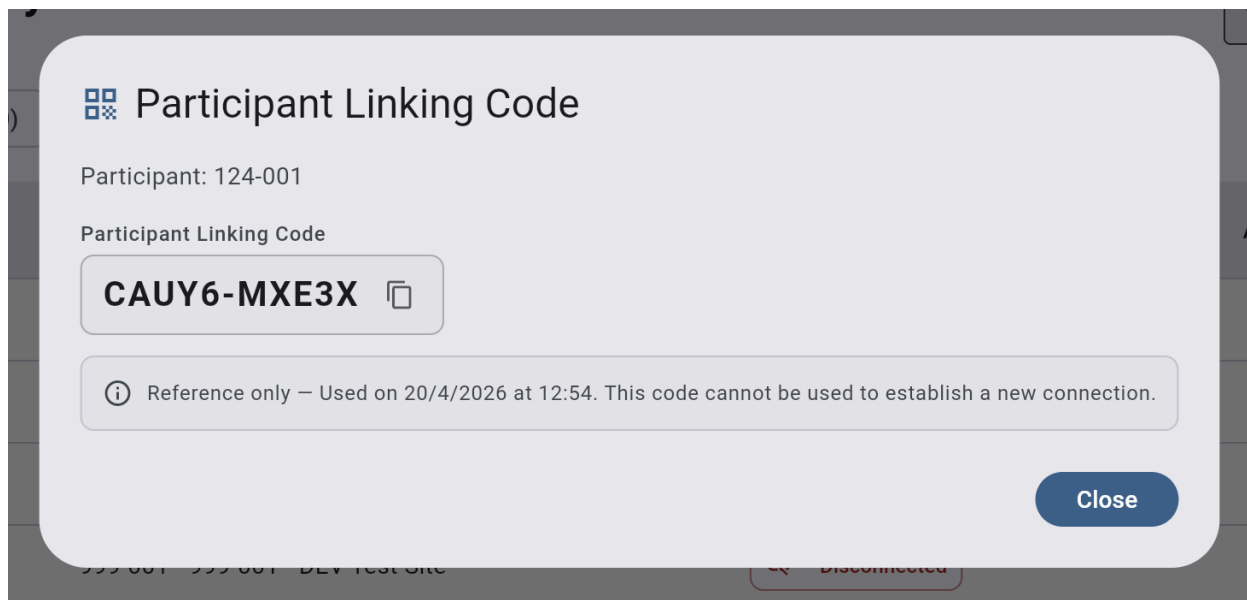
Dismissal: **Close**.

Reference: GUI-p03001.

Pending Status



After Participant was linked to the Mobile Application



7.2.3 Manage Questionnaires Modal

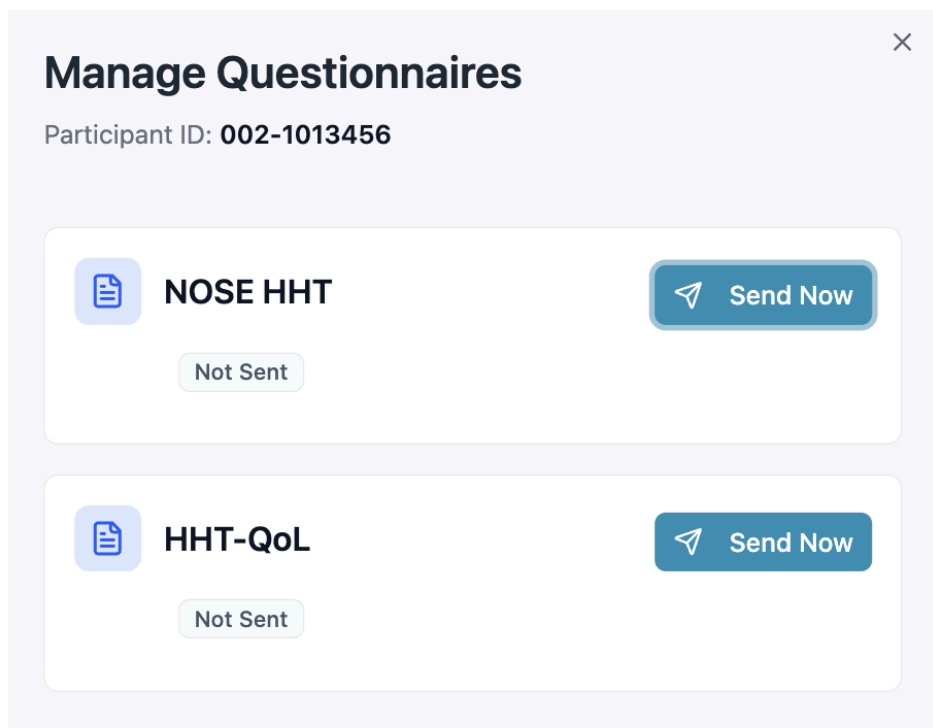
Trigger: **Study Coordinator** selects **Manage Questionnaires** for a **Participant** with **Trial Active** status.

Target identification: **Participant** ID in the modal header.

Displays: One **Questionnaire Card** per **Questionnaire Type** enabled for the deployment. Each card shows the **Questionnaire Type** name and current **Questionnaire Status**, with content and actions varying by status and **Cycle** state (e.g., **Send Now**, **Start Next Cycle**, **Finalize**, call back).

Dismissal: Close action.

Reference: GUI-CAL-p00006.



The screenshot shows a modal titled "Manage Questionnaires" with a close button (X) in the top right corner. Below the title, the "Participant ID: 002-1013456" is displayed. The modal contains two questionnaire cards. The first card is for "NOSE HHT" and shows a "Not Sent" status with a "Send Now" button. The second card is for "HHT-QoL" and also shows a "Not Sent" status with a "Send Now" button. Each card has a document icon on the left.

Questionnaire Type	Status	Action
NOSE HHT	Not Sent	Send Now
HHT-QoL	Not Sent	Send Now

7.3 Dialogs

7.3.1 Deactivate User Account — Reason Dialog (Free Text)

Trigger: **Administrator** initiates deactivation of a **User Account** from the **Active Users** tab. *Target identification:* **Full Name**.

Consequence summary: The **User Account** will be deactivated and the user will no longer be able to access the **System**.

Pattern: Reason Dialog — Free Text (see §4.1).

Reference: REQ-p20031, GUI-p00031.

Deactivate User Account

You are about to deactivate the account for "Dr. Sarah Johnson".

Effects of this action:

- Terminate all active sessions immediately
- Prevent the user from logging in
- Move the user to the Inactive Users tab

This action can be reversed by reactivating the user.

Reason for deactivation *

0/100

CancelConfirm

7.3.2 Reactivate User Account — Reason Dialog (Free Text)

Trigger: **Administrator** initiates reactivation of a **User Account** from the **Inactive Users** tab.

Target identification: **Full Name**.

Consequence summary: The **User Account** will be set to **Pending Activation** and the user will be required to complete the activation workflow before regaining access.

Pattern: Reason Dialog — Free Text (see §4.1).

Reference: REQ-p20032, GUI-p00032.

Reactivate User Account

You are about to reactivate the account for "**Dr. Michael Chen**".

Effects of this action:

- Restore previously assigned roles and site assignments
- Set account status to Pending Activation
- Send a new activation email to the user
- Require the user to set a new password and complete 2FA setup

The user will not be able to log in until they complete the activation workflow.

Reason for reactivation *

0/100

CancelConfirm

7.3.3 Link Participant — Confirmation Dialog

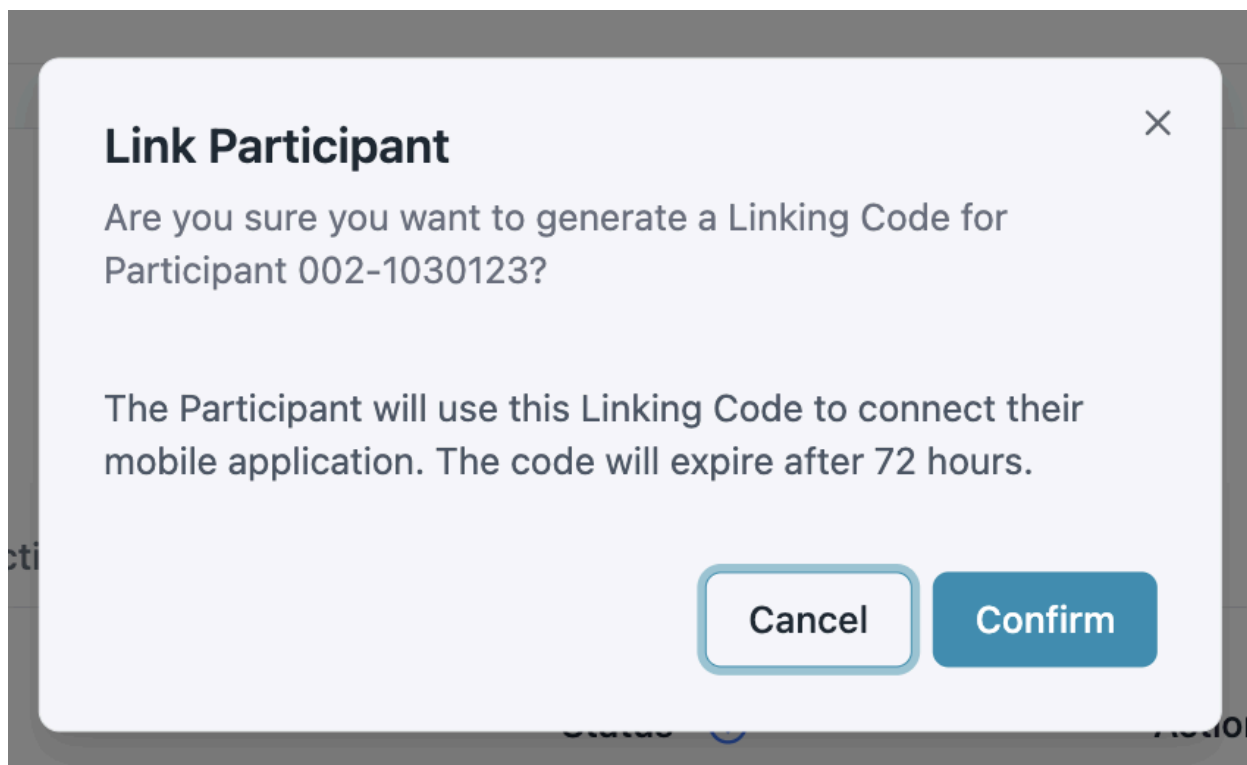
Trigger: **Study Coordinator** initiates **Link Participant** for a **Participant** with **Not Connected** status.

Target identification: **Participant ID**.

Consequence summary: A **Mobile Linking Code** will be generated for the **Participant** and will expire after the configured duration.

Pattern: Confirmation Dialog (see §4.1).

Reference: REQ-p70009, GUI-p03001.



7.3.4 Linking Code Generated — Acknowledgement Dialog

Trigger: **Study Coordinator** confirms generation of a **Mobile Linking Code** from the Link Participant Confirmation Dialog (§7.3.3).

Target identification: **Participant** ID.

Displays: Generated **Mobile Linking Code** with a **Copy** action, and remaining time until expiry.

Outcome: On dismissal, the **Participant's Status Badge** updates to **Pending**.

Pattern: Acknowledgement Dialog (see §4.1).

Reference: GUI-p03001.

Linking Code Generated

The Linking Code has been generated successfully. Share it with the Participant to connect their Mobile Application.

Participant: 001-1028901

IHVI7-SSS36



Expires in 3 days, 0 hours

OK

7.3.5 Start Trial — Confirmation Dialog

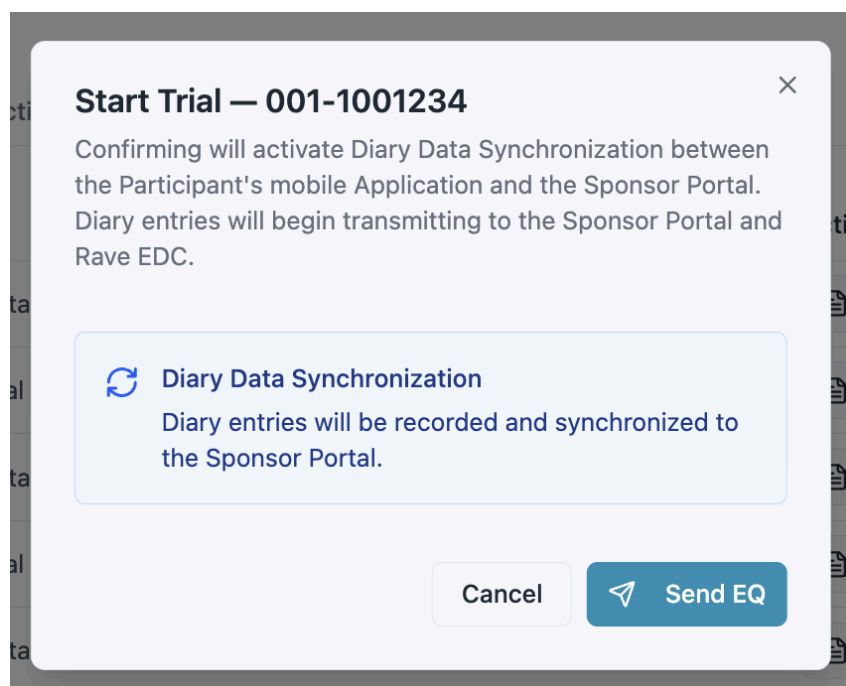
Trigger: **Study Coordinator** selects **Start Trial** for a **Participant** with **Linked - Awaiting Start** status.

Target identification: **Participant ID**.

Consequence summary: **Diary Data Synchronization** will be activated, transmitting diary entries to the **Sponsor Portal** and **Rave EDC**. The **Participant's** status will update to **Trial Active**.

Pattern: Confirmation Dialog (see §4.1).

Reference: REQ-CAL-p00022, GUI-CAL-p00005.



7.3.6 Disconnect Participant — Reason Dialog

Trigger: **Study Coordinator** selects **Disconnect Participant** from the **Participant Actions Modal** for a **Participant** with **Linked - Awaiting Start** or **Trial Active** status.

Target identification: **Participant ID**.

Consequence summary: Data synchronization between the **Mobile Application** and the **Sponsor Portal** will stop. All **Participant** data will be preserved.

Reason options: Device Issues, Technical Issues, Other.

Pattern: Reason Dialog — Predefined List (see §4.1).

Reference: REQ-p70010, REQ-CAL-p00020.

Disconnect Participant

Participant ID: 002-1006789

Effects of this action:

- Stop data synchronization between the Participant's Mobile Application and the Sponsor Portal
- Continue applying sponsor-specific rules to the Participant's Mobile Application
- Preserve all Participant data and history

This action can be reversed by reconnecting the Participant.

Reason for disconnection *

Select a reason

Cancel

Confirm

7.3.7 Reconnect Participant — Reason Dialog (Free Text)

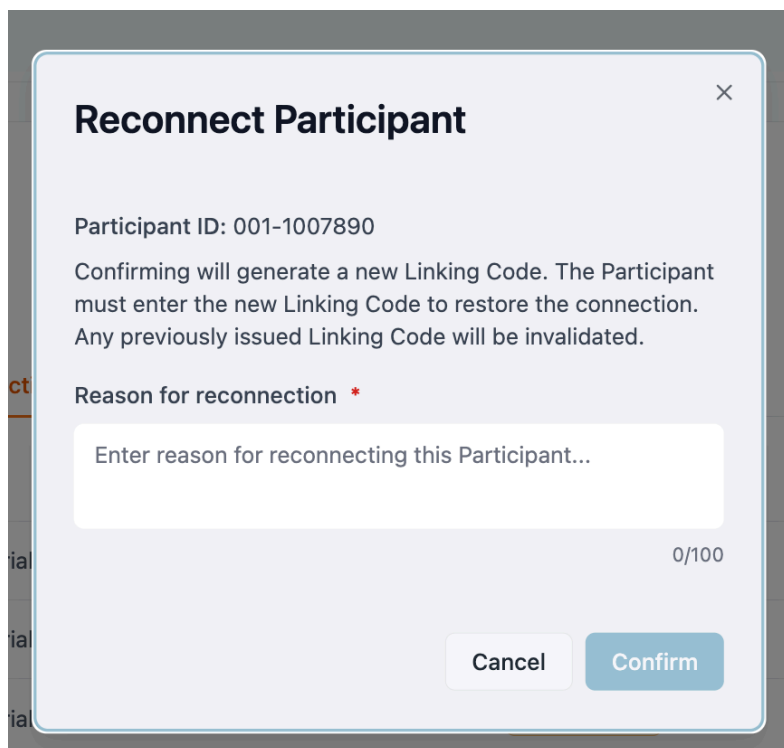
Trigger: **Study Coordinator** selects **Reconnect Participant** from the **Participant Actions Modal** for a **Participant** with **Disconnected** status.

Target identification: **Participant ID**.

Consequence summary: A new **Mobile Linking Code** will be generated. The **Participant** must enter the new code to restore the connection.

Pattern: Reason Dialog — Free Text (see §4.1).

Reference: REQ-p70011.



The image shows a modal dialog box titled "Reconnect Participant" with a close button (X) in the top right corner. The dialog contains the following text: "Participant ID: 001-1007890", "Confirming will generate a new Linking Code. The Participant must enter the new Linking Code to restore the connection. Any previously issued Linking Code will be invalidated.", and a required field "Reason for reconnection *" with a red asterisk. Below this is a text input field with the placeholder "Enter reason for reconnecting this Participant..." and a character count "0/100" on the right. At the bottom are two buttons: "Cancel" and "Confirm".

7.3.8 Mark as Not Participating — Reason Dialog (Predefined)

Trigger: Study Coordinator selects **Mark as Not Participating** for a **Participant** with **Disconnected** status.

Target identification: **Participant** ID.

Consequence summary: Sponsor-specific rules will no longer be applied to the **Participant's Mobile Application**. *Reason options:* Subject Withdrawal, Death, Protocol Treatment/Study Complete, Other.

Pattern: Reason Dialog — Predefined List (see §4.1).

Reference: REQ-p70017, REQ-CAL-p00064.

Mark Participant as Not Participating

Participant ID: 001-1023456

Effects of this action:

- Stop applying sponsor-specific rules to the Mobile Application
- Preserve all Participant data and history

This action can be reversed by reactivating the Participant.

Reason for marking as not participating *

Select a reason

Cancel

Confirm

7.3.9 Reactivate Participant — Reason Dialog (Free Text)

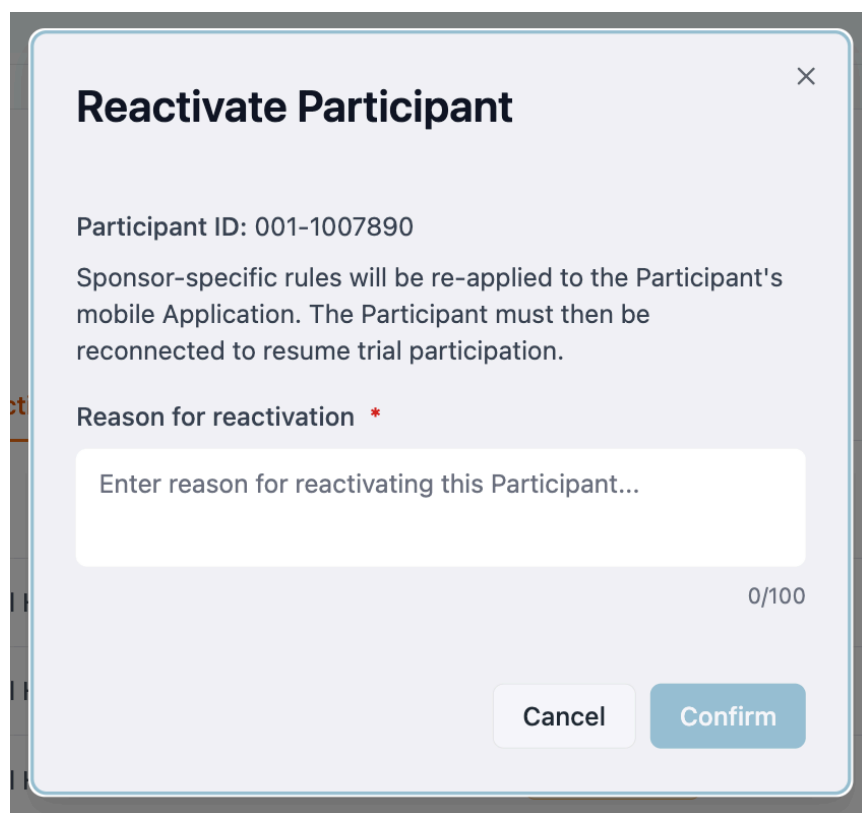
Trigger: **Study Coordinator** selects **Reactivate Participant** from the **Participant Actions Modal** for a **Participant** with **Not Participating** status.

Target identification: **Participant ID**.

Consequence summary: Sponsor-specific rules will be re-applied to the **Participant's Mobile Application** and the standard reconnection workflow will become available.

Pattern: Reason Dialog — Free Text (see §4.1).

Reference: REQ-p70016.



Reactivate Participant ×

Participant ID: 001-1007890

Sponsor-specific rules will be re-applied to the Participant's mobile Application. The Participant must then be reconnected to resume trial participation.

Reason for reactivation *

Enter reason for reactivating this Participant...

0/100

Cancel Confirm

7.3.10 Delete Record — Reason Dialog — Predefined List

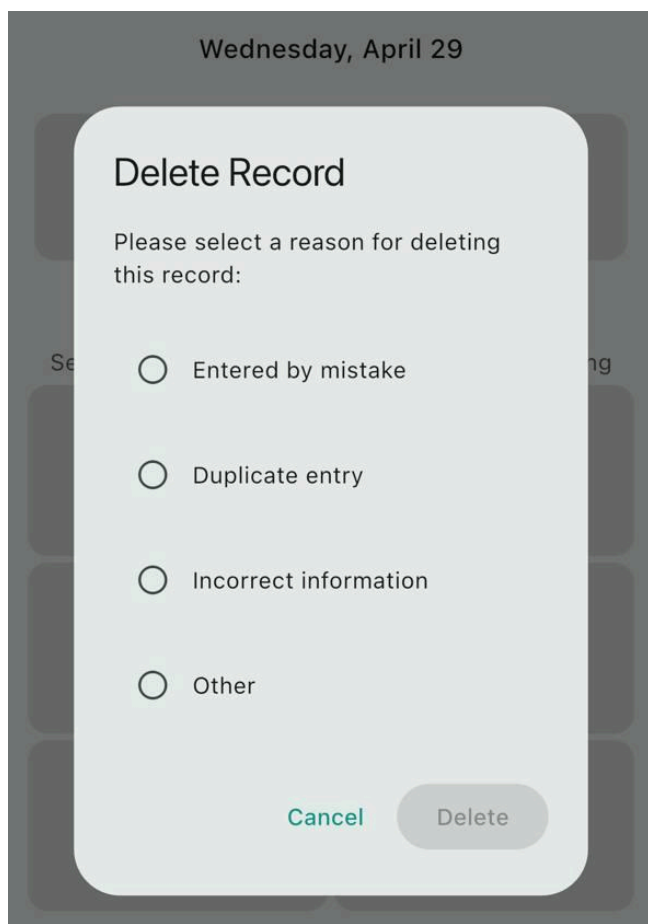
Trigger: **Participant** initiates the delete action on an **Epistaxis Event**, either during the recording flow or from event history.

Consequence summary: The **Epistaxis Event** will be removed from the **Participant's** diary.

Reason options: Entered by mistake, Duplicate entry, Incorrect information, Other.

Pattern: Reason Dialog — Predefined List (see §4.1).

Reference: GUI-p00003.



Wednesday, April 29

Delete Record

Please select a reason for deleting this record:

- ☐ Entered by mistake
- ☐ Duplicate entry
- ☐ Incorrect information
- ☐ Other

[Cancel](#) [Delete](#)

7.3.11 Post-Submission Acknowledgement Dialog

Trigger: **Participant** confirms **Submission** of a **Portal-Sent Questionnaire**.

Consequence summary: Confirms the **Questionnaire** has been submitted and informs the **Participant** that the **Study Coordinator** will review the responses.

Pattern: Acknowledgement Dialog (see §4.1).

Reference: GUI-p00001.



Questionnaire Submitted

Your questionnaire has been submitted successfully. Your
Study Coordinator will review your responses.

OK

7.3.12 Session Expiry Dialog

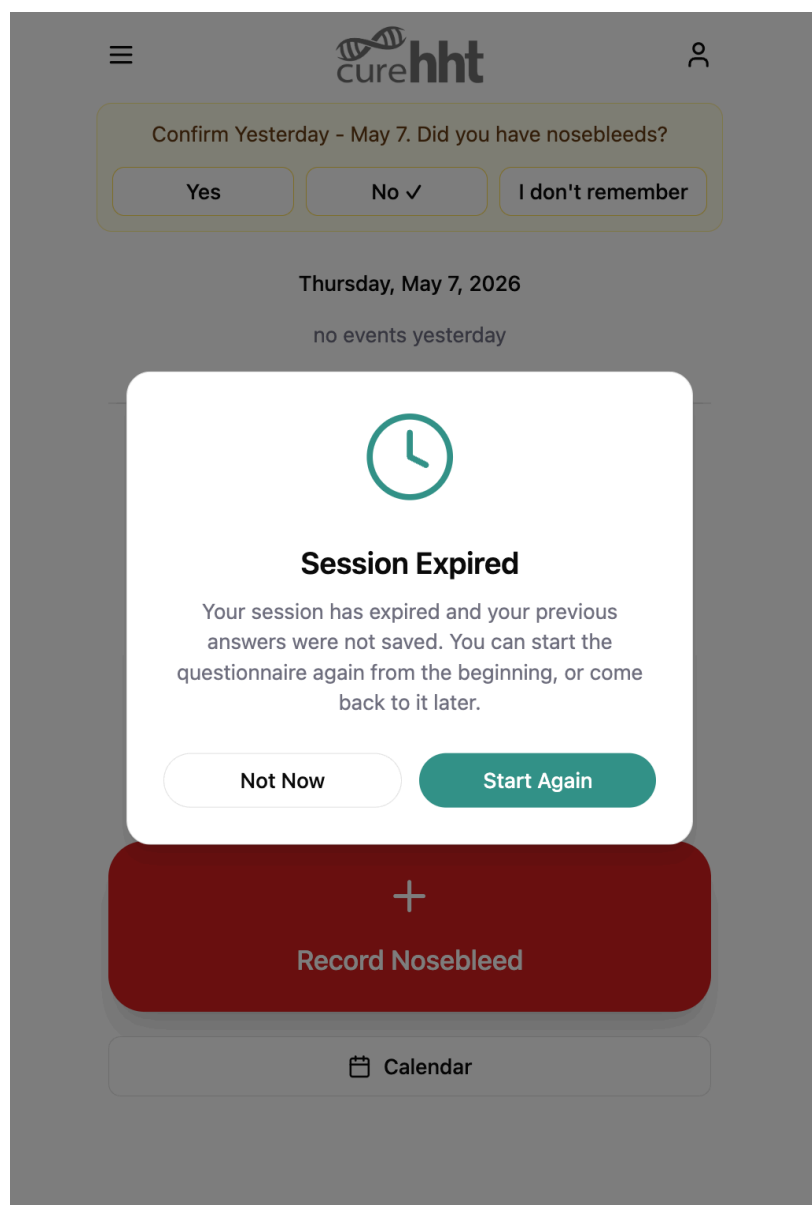
Trigger: **Participant** opens a **Portal-Sent Questionnaire** that has reached **Session Expiry**.

Consequence summary: Informs the **Participant** that the session has expired and previous answers were not saved.

Actions: **Start Again** (begin the **Questionnaire** from the **Preamble**), **Not Now** (return to **Main Screen**).

Pattern: Acknowledgement Dialog (see §4.1).

Reference: REQ-p01073, GUI-p00004.



7.3.13 Questionnaire Finalization Dialog

Trigger: Study Coordinator selects **Finalize** on a **Questionnaire Card** with **Delivery Failed** or **Ready to Review** status.

Consequence summary: Finalizes the Questionnaire — locks all Participant responses permanently, calculates the score, transmits it to Rave EDC, and prevents the Participant from editing their answers in the Mobile Application.

Actions: Finalize Questionnaire (apply finalization with the selected Cycle), Cancel (close the dialog with no changes).

Inputs: Cycle dropdown — selectable values are the Current Cycle N Day 1 value, End of Treatment, and End of Study.

Cross-flow: When a Terminal Cycle value (End of Treatment or End of Study) is selected, confirming opens the Terminal Cycle Warning Dialog before finalization is applied.

Pattern: Confirmation Dialog (see §4.1).

Reference: REQ-CAL-p00023 I–M, GUI-CAL-p00007 A–F.

Finalize Questionnaire?

Participant ID: 002-1009012

Are you sure you want to finalize the NOSE HHT Questionnaire?

Cycle

Cycle 4 Day 1

 This questionnaire will be finalized as: **Cycle 4 Day 1**

Effects of this action:

- Finalize this Questionnaire
- Calculate the score and send it to EDC
- Lock all Participant responses permanently

After finalization, the Participant cannot edit or update their answers in the Daily Diary app.

Cancel

Confirm

7.3.14 Terminal Cycle Warning Dialog

Trigger: **Study Coordinator** confirms finalization of a **Questionnaire** with a **Terminal Cycle** value (**End of Treatment** or **End of Study**) selected in the **Finalization Dialog**.

Target identification: **Participant ID**, **Questionnaire Type**, **Terminal Cycle** being assigned.

Consequence summary: The **Participant's** answers will be locked, the score will be transmitted to **Rave EDC**, and the **Questionnaire Type** will be closed for the **Participant** so no further **Questionnaires** of that type may be sent.

Cancellation behavior: On cancel, the **Questionnaire** remains unchanged and the **Study Coordinator** is returned to the **Finalization Dialog**.

Pattern: Confirmation Dialog (see §4.1).

Reference: GUI-CAL-p00007.

Finalize Terminal Cycle

Participant ID: 002-1006789

Finalizing Cycle 1 — End of Study for HHT-QoL.

Effects of this action:

- Lock the Participant's answers permanently
- Calculate the Questionnaire score and transmit it to Rave EDC
- Close this Questionnaire Type for the Participant

After finalization, no further HHT-QoL Questionnaires can be sent to this Participant.

Cancel

Confirm

7.3.15 Call Back Questionnaire — Reason Dialog (Free Text)

Trigger: **Study Coordinator** initiates the call back action on a **Questionnaire** with **Sent**, **Delivery Failed**, or **Ready to Review** status.

Target identification: **Participant ID**, **Questionnaire Type**.

Consequence summary: The **Questionnaire** status will be set to **Not Sent**.

Pattern: Reason Dialog — Free Text (see §4.1).

Reference: REQ-CAL-p00023, GUI-CAL-p00006.

Call Back Questionnaire

Participant ID: 002-1013456

Calling back the NOSE HHT Questionnaire will set its status to Not Sent.

Reason for call back *

Enter reason for calling back this Questionnaire...

0/100

Cancel

Confirm

7.3.16 Call Back Notice

Trigger: A **Portal-Sent Questionnaire** assigned to the **Participant** has been called back by the **Study Coordinator**.

Consequence summary: Informs the **Participant** that the **Questionnaire** is no longer active and any answers entered will not be saved.

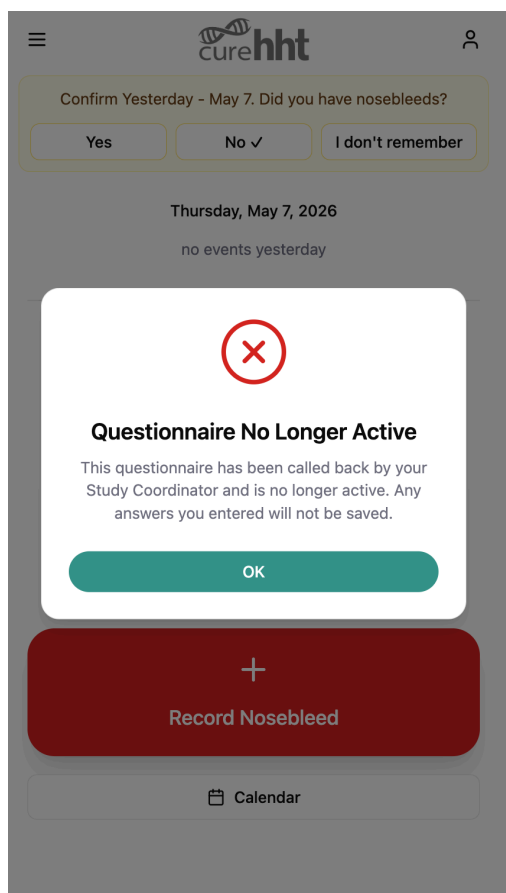
Presentation contexts:

- The **Portal-Sent Questionnaire** is called back while the **Participant** has it open.
- The **Participant** opens the **Mobile Application** after a call back has occurred.
- The **Participant** attempts to submit a **Portal-Sent Questionnaire** that has been called back while offline.

Behavior: Cannot be dismissed by tapping outside its bounds. On acknowledgement, the **Participant** is returned to the **Main Screen** and the corresponding **Questionnaire Task** is removed from the **Task List**.

Pattern: Acknowledgement Dialog (see §4.1).

Reference: GUI-CAL-p07002.



7.3.17 Successful Linking Confirmation

Trigger: A **Participant** successfully submits a valid **Mobile Linking Code** on the **Join the Study** screen.

Consequence summary: Confirms to the **Participant** that the device has been linked to the study and transitions the **Participant** from personal use mode into linked use mode.

Behavior: On acknowledgement, the **Participant** is navigated to the **User Profile** screen.

Pattern: Acknowledgement Dialog (see §4.1).



Device Linked Successfully

Your device has been successfully linked to the study.

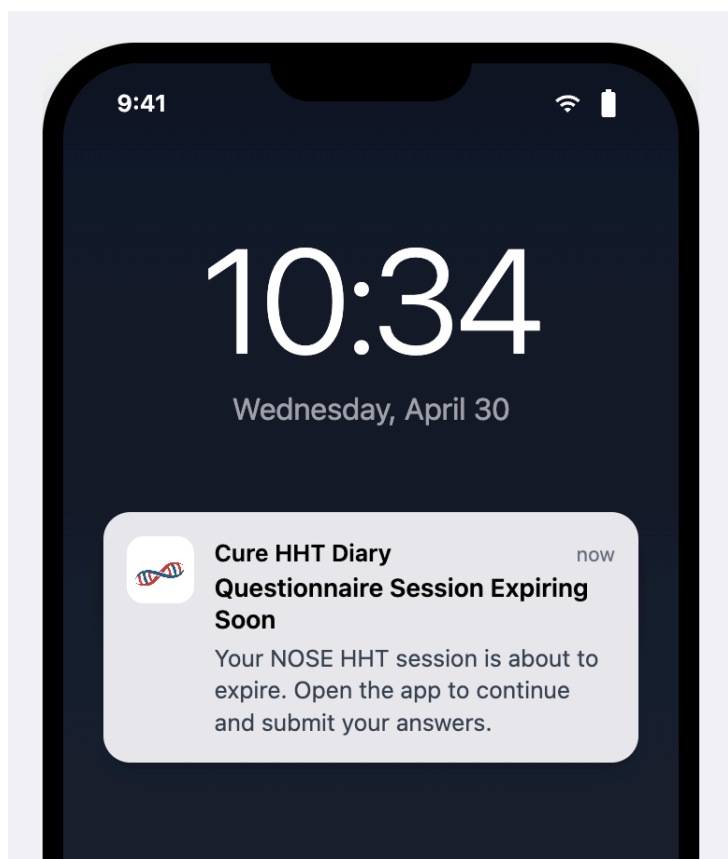
OK

Reference: GUI-p05015.

7.4 Notifications

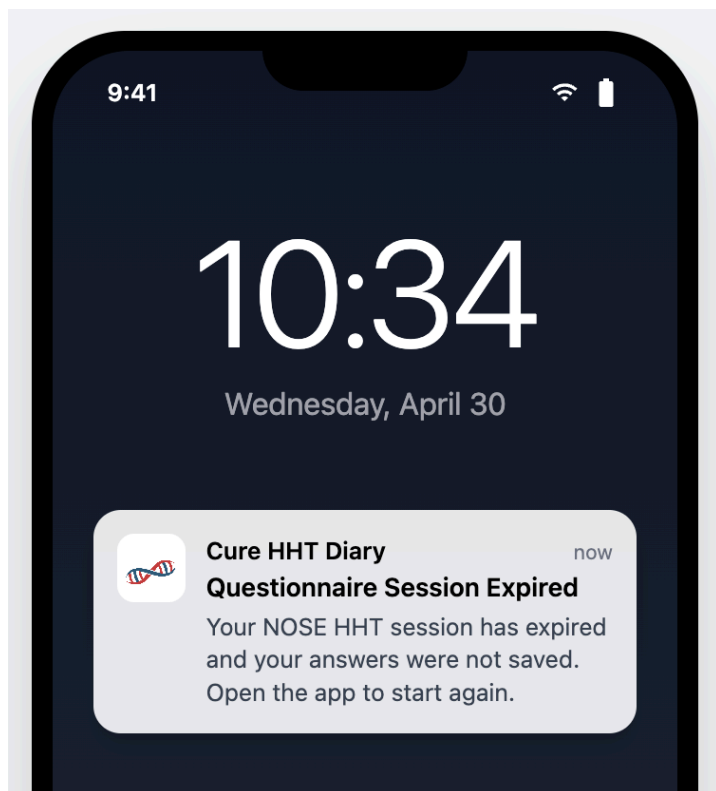
7.4.1 Timeout Warning Notification

A push notification delivered to the Participant when a configured Questionnaire Session Timeout is approaching expiry. Tapping the notification opens the questionnaire in the Mobile Application. The wording is sponsor-configurable per Questionnaire definition; the default text is *["Your [Questionnaire Display Name] session is about to expire. Open the app to continue and submit your answers."]*



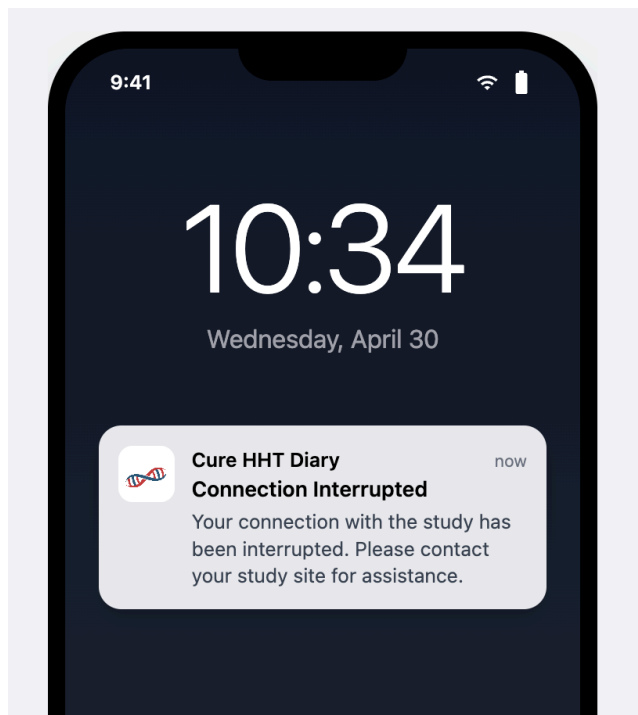
7.4.2 Session Expiry Notification

A push notification delivered to the Participant when the configured Session Timeout has been exceeded for an in-progress Questionnaire and the Participant's previously entered answers have been discarded. Tapping the notification opens the Mobile Application; if the Participant returns to the Questionnaire, the Session Expiry Dialog is displayed. The wording is sponsor-configurable per Questionnaire definition; the default text is *["Your [Questionnaire Display Name] session has expired and your answers were not saved. Open the app to start again."]*



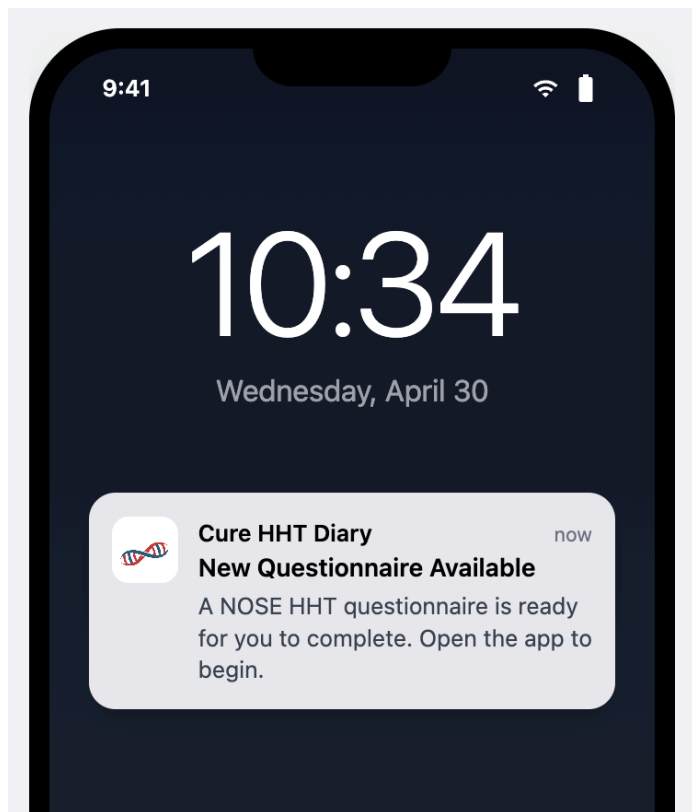
7.4.3 Disconnection Notification

A persistent, non-dismissible in-app notification displayed in the System Notice Area at the top of the Main Screen when the Participant's status is Disconnected. The notification persists until the Participant is reconnected to the Sponsor Portal. The wording is sponsor-configurable; the default text is *["Your connection with the study has been interrupted. Please contact your study site for assistance."]*



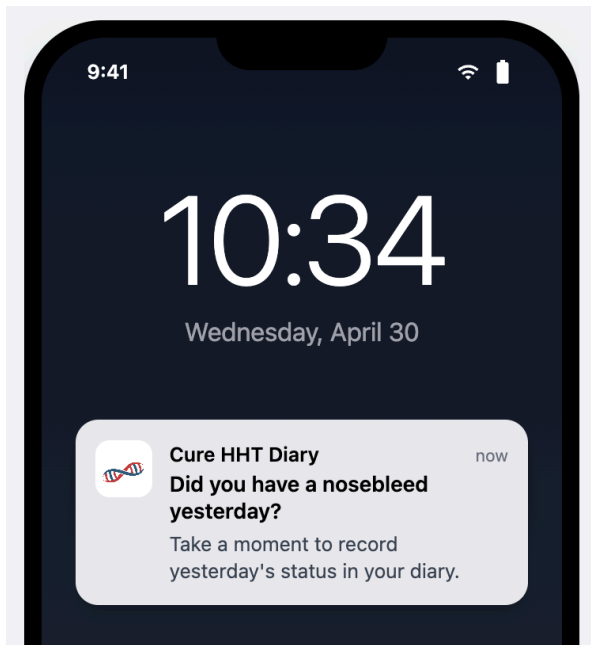
7.4.4 Portal-Sent Questionnaire Notification

A push notification delivered to the Participant when a Study Coordinator sends a Portal-Sent Questionnaire from the Sponsor Portal. Tapping the notification opens the Mobile Application; the Participant accesses the Questionnaire from the Questionnaire Task on the Main Screen. The wording is sponsor-configurable; the default text is *"A [Questionnaire Display Name] questionnaire is ready for you to complete. Open the app to begin."*



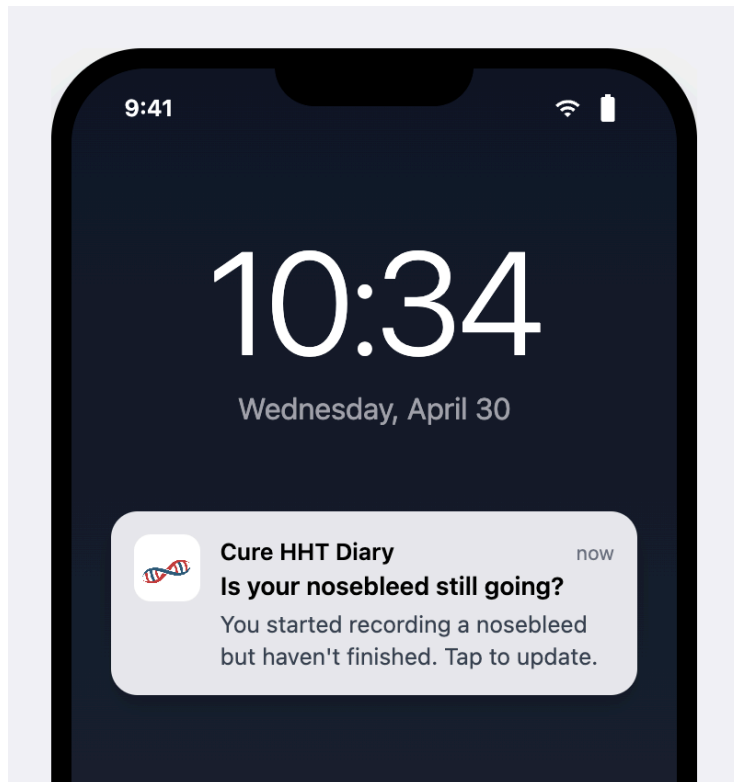
7.4.5 Yesterday Entry Reminder Notification

A push notification delivered to the Participant at the configured Reminder Time when no Daily Status has been recorded for the previous calendar day. Tapping the notification opens the Mobile Application; the Participant accesses the Yesterday Reminder Task on the Main Screen. The wording is sponsor-configurable; the default text is *["Did you have a nosebleed yesterday? Take a moment to record yesterday's status in your diary."]*



7.4.6 Ongoing Epistaxis Event Reminder

A push notification delivered to the Participant when an Incomplete Record has not been interacted with for the configured Reminder Interval. Tapping the notification opens the Mobile Application; the Participant returns to the Incomplete Record to either complete it or confirm the event is still ongoing. The wording is sponsor-configurable; the default text is *["Is your nosebleed still going? You started recording a nosebleed but haven't finished. Tap to update."]*



7.4.7 Historical Gap Reminder

A push notification delivered to the **Participant** at the configured Reminder Time when one or more Historical Gaps exist within the editable window. Tapping the notification opens the Mobile Application; the Participant accesses the missing days via the Calendar. The wording is sponsor-configurable; the default text is [*"You have one day without a recorded entry. Tap to review and complete your diary."*]

